

Time-reversal technique for pipeline defect detection

Muhammad Waqar^{*}, Moez Louati, Mohamed S. Ghidaoui

Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Sai Kung, New Territories, Hong Kong

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ABSTRACT

In this paper, time-reversal (TR) technique is proposed for the detection of discrete defects (e.g., leaks and blockages) in pressurized fluid-filled pipelines using active transient waves. The proposed technique is based on the one-dimensional wave equation and decouples the search for the location from that of the size of the defect. Another key feature of this method is that the measured pressure head signal can be truncated so as to eliminate undesirable reflections from unknown or uncertain parts of the pipe system so as not to compromise the accuracy and robustness of the detection. The proposed method is tested through numerical simulations for a single leak, multiple leaks, and multiple mixed-type of defects (i.e., leaks and blockages) in elastic as well as visco-elastic pipe systems. The proposed method is also validated through laboratory experiments wherein leaks and blockages of various sizes are considered. The sensitivity analysis of the method with respect to wave speed and signal-to-noise ratio (SNR) is also performed. In all cases, the method has shown accurate results in terms of localization, classification, and size estimation of the defects even for SNR as low as 0 dB.

1. Introduction

Freshwater pipelines are susceptible to defects such as leaks, blockages, malfunctioning devices, etc. Such defects are responsible for loss of water, energy and financial resources (Colombo and Karney, 2002). According to the World Bank report, leaks in freshwater pipelines are responsible for about 32 billion cubic meters of water loss per year worldwide (World Bank, 2016, pp. 43). To place this number in some perspective, this amount of lost water could fill 12,800 Olympic size swimming pools. To control such huge losses, development of defect detection technologies is the pragmatic solution recommended by many organizations concerned with water resources such as UNESCO (UNESCO, 2020) and Asian Development Bank (Strazzabosco, 2020).

In past three decades, immense attention has been given by hydraulic researchers to develop transient-based defect detection methods (TBDDMs; Che et al., 2021). These methods are based on the principle that injected transient wave propagates through water column and interacts with all physical features of the pipeline including defect (if present). The transient response of the system is recorded through pressure transducers which are mounted at several locations along the pipe. The recorded data is then used for pipeline condition assessment.

In literature, a variety of TBDDMs have been proposed to extract the defect information from the recorded transient signals. Some of these methods directly analyze the measured signals using the wave reflectometry principles (e.g., Brunone, 1999; Ferrante et al., 2007;

Meniconi et al., 2015). A number of closed-form analytical solutions are also proposed in literature to extract defect information for simple systems such as reservoir-pipe-reservoir, reservoir-pipe-valve, etc. (see e.g., Mpesha et al., 2002; Lee et al., 2005; Sattar and Chaudhry, 2008; Duan et al., 2011; Louati et al., 2020, 2018, 2017). A great deal of attention has been devoted to the development of inverse techniques to estimate the parameters of defects based on matching the measured and modeled transient response (e.g., Liggett and Chen, 1994; Vitkovsky et al., 2001; Kapelan et al., 2003; Covas and Ramos, 2010; Soares et al., 2011; Wang and Ghidaoui, 2018b).

In theory, the various inverse methods are efficient given that: (i) the defect type (leak/blockage) and number of defects are known a priori (Puust et al., 2010), (ii) the topology of the system and its physical and geometrical properties are well-known and fully incorporated in the hydraulic model (Stephens, 2008; Waqar et al., 2021), and (iii) the signal-to-noise ratio (SNR) is large (Wang and Ghidaoui, 2018b; Wang et al., 2020b). In real-world, however, most of the foregoing conditions are not met. Consequently, the application of existing TBDDMs (e.g., the time-domain reflectometry, gradient-based optimization algorithms, and/or resonant peak frequency pattern methods) is restricted to simple pipelines with large SNR and known defect type.

The time reversal-based (TR) techniques are known to be robust to noise and model uncertainties (Tolstoy, 1993; Schmidt et al., 1990; Kuperman et al., 1998; Edelmann et al., 2002; Kuperman and Jackson,

^{*} Corresponding author.

E-mail address: mwaqar@connect.ust.hk (M. Waqar).

2002; Grigoropoulos et al., 2022). They have been extensively developed in various fields for defect detection purposes such as (i) in medical imaging to detect and destroy brain tumors & kidney stones (Fink et al., 2003); (ii) in structures to detect fractures (Park et al., 2007); (iii) in geophysics to localize earthquake source (Larmat et al., 2010); (iv) in underwater acoustics to assist in long-distance communications (Roux and Fink, 2000; Edelmann et al., 2005); and (v) in the study of Phononic Crystals and meta-materials to control and manipulate waves (Lemoult et al., 2016; Liu et al., 2020; Li et al., 2021; Ahmed et al., 2022; Waqar et al., 2023). In the context of pipeline hydraulics, the authors of this paper recently showed that the time reversal of transient waves works efficiently, allowing the waves to refocus on their source location even in the presence of waveguide damping such as steady friction (SF), unsteady friction (UF) and visco-elastic (VE) effects of the polyethylene pipes (Waqar et al., 2020, 2022). Thus, the TR technique has a great potential to be used as a tool for pipeline defect detection which is the primary objective of this paper.

The remainder of the paper is organized as follows. First, the TR technique for defect detection in pipelines is developed wherein analytical models in both time and frequency domains are provided. In this paper, only discrete defects such as leaks or discrete blockages are considered. Second, numerical examples of single leak, multiple leaks and mixed-type of defects (e.g., leak and discrete blockage) are considered in a single transmission-like pipeline. Third, the proposed method is validated through laboratory experiments under different SNR values ranging between 10 to -7 dB. Finally, the conclusions of this study are stated.

1.1. Analytical model

Consider a visco-elastic (VE) pipe system as shown in Fig. 1. The system is comprised of a valve at the downstream end ($x = x_S$) and a wave reflector at $x = \chi$ which scatters back part of the incident wave. In this paper, two kinds of reflectors are considered: (i) mass source reflector (e.g., leak or active illegal branch), and (ii) momentum source reflector (e.g., discrete blockage or partially closed in-line valve, etc.). Let the injected signal be $\delta(x - x_S, t - t_S)$, where x_S is the location of the source and t_S denotes the timing of the source. The impulse response of the system is derived in Appendix A and the final expression is given by Eq. (A.29) as

$$\tilde{h}(\omega, x_R|\chi) = Z\tilde{q}_S \exp[-ik(x_R - x_S)] + \tilde{A}^\chi \tilde{G}(\omega, \chi) \quad (1)$$

where $\tilde{h}$ is the impulse response function in the frequency domain, ω is the angular frequency, x_R is the receiver location, k is the wave number (given by Eq. (A.7)), Z is the characteristic impedance of the pipe (given by Eq. (A.10)), $\tilde{q}_S = 1$ represents the unit strength of the point source in the frequency domain and has the dimension of $[L^3]$,

$$\tilde{G}(\omega, \chi) = \exp[-2ik(\chi - x_S)] \exp[ik(x_R - x_S)] \quad (2)$$

is the propagation function from reflector location χ to measurement location x_R ,

$$\tilde{A}^\chi = \begin{cases} \tilde{A}_{\text{mass}}^\chi = \frac{-|Z|^2}{Z + 2Z_{\text{mass}}^\chi} \tilde{q}_S & \text{for mass source reflector (i.e., a leak)} \\ \tilde{A}_{\text{mom}}^\chi = \frac{Z Z_{\text{mom}}^\chi}{2Z + Z_{\text{mom}}^\chi} \tilde{q}_S & \text{for momentum source reflector} \\ & \text{(i.e., a blockage)} \end{cases} \quad (3)$$

is the amplitude of the reflection from the reflector and has the dimension of $[L.T]$ in the frequency domain, Z_{mass}^χ is the impedance of the mass source reflector (given by Eq. (A.19)), and Z_{mom}^χ is the impedance of the momentum source reflector (given by Eq. (A.26)). Note that k and Z are complex which include linearized steady-state friction (SF), unsteady friction (UF) and VE effects of the pipe wall. All

these damping effects are accounted for in the characteristic impedance of the pipe wall (denoted by Z and defined in Eq. A10 in Appendix A). In this paper, the UF is modeled using Vardy–Brown model (Vardy and Brown, 1995) and VE effects are incorporated using the Kelvin–Voigt (KV) model (Covas et al., 2005). Let

$$\Delta\tilde{h} = \tilde{h} - Z\tilde{q}_S \exp[-ik(x_R - x_S)]$$

and $x_R = x_S$, then Eq. (1) becomes

$$\Delta\tilde{h}(\omega, x_R|\chi) = \tilde{A}^\chi \tilde{G}(\omega, \chi) \quad (4)$$

For an ideal system where there is no damping, $Z = a_0/(gA)$, $k = \omega/a_0$, where a_0 is the wave speed (given by Eq. (A.3)), A is the cross-sectional area of the pipe, and g is the gravitational acceleration constant. Then, the inverse Fourier transform of Eq. (4) results

$$\Delta h(t, x_R|\chi) = A^\chi G(t, \chi), \quad (5)$$

where $\Delta h(t) = \mathcal{F}^{-1}[\Delta\tilde{h}(\omega)]$, $\mathcal{F}^{-1}$ denotes the inverse Fourier transform, A^χ denotes the amplitude of the reflected wave in time domain,

$$G(t, \chi) = \mathcal{F}^{-1}[\tilde{G}(\omega, \chi)] = \delta\left[t - \frac{2(\chi - x_S)}{a_0}\right] \quad (6)$$

is the propagation function, and x_S is the location of transient source. While it may be tempting to think of $G(t)$ as the Green's function, this is not the case in this paper. $G(t)$ is a generic propagation function which accounts for the wave propagation between source (valve) and the defect. In practice, $\delta(x - x_S, t - t_S)$ is realized by injecting a pulse-type waveform by rapidly closing and opening the initially open valve. As such, the response of the system subjected to a pulse-type waveform has a finite bandwidth and is represented by

$$\Delta H(t, x_R|\chi) = q_{in}(t) * \Delta h(t, x_R|\chi) \quad (7)$$

where $\Delta H(t)$ is the pressure head response of the system, $\Delta h(t)$ is the impulse response function, and $q_{in}(t)$ is the non-dimensional pulse-type forcing function whose magnitude is equals to the Joukowski pressure head (Wylie et al., 1993). Additionally, since noise is inevitably present in the measured signals, $\Delta H(t)$ is contaminated with measurement noise n such that

$$\Delta H_m(t, x_R|\chi) = \Delta H + n \quad (8)$$

where ΔH_m denotes the measured pressure head response and n is assumed to follow Gaussian distribution with zero mean and fixed variance σ^2 (i.e., $n \sim \mathcal{N}(0, \sigma^2)$). The objective of this paper is to extract the defect characteristics (i.e., χ and A^χ) from ΔH_m using the time-reversal technique.

1.2. Time-reversal technique

The standard time-reversal technique is a two stage process. In stage 1: a point source is injected into the system and the response of the system is measured. For a pipe system in Fig. 1, the measured pressure head response is denoted by $\Delta H_m(t, x_R|\chi)$ (i.e. Eq. (8)). In stage 2, the measured response is first reversed (flipped) in time such that $\Delta H_m(t, x_R|\chi)$ becomes $\Delta H_m(-t, x_R|\chi)$. This time-reversed signal is then re-emitted back to the medium and the response of the system is measured again. For leak detection application, this second stage is performed in the computer and not in the pipeline system wherein the response is predicted by taking the convolution product of the input signal $\Delta H_m(-t, x_R|\chi)$ with the theoretical impulse response function of the system $\Delta h(t, x_R|\hat{\chi})$, where $\hat{\chi}$ is the estimated value of χ . It is shown in Appendix B that the time-reversed signal $\Delta H_m(-t, x_R|\chi)$ acts as an optimal filter during the second stage of TR which maximizes the signal-to-noise ratio only when $\hat{\chi} = \chi$. Hence, the optimal estimate for leak location $\hat{\chi}$ can be obtained by maximizing the following objective function B :

$$\hat{\chi} = \arg \max_{\hat{\chi}} B = \arg \max_{\hat{\chi}} \int_{-\infty}^{\infty} \frac{\mathcal{G}(\tau, x_R|\hat{\chi})}{\|\mathcal{G}(\tau, x_R|\hat{\chi})\|} \cdot \Delta H_m(\tau, x_R|\chi) d\tau. \quad (9)$$

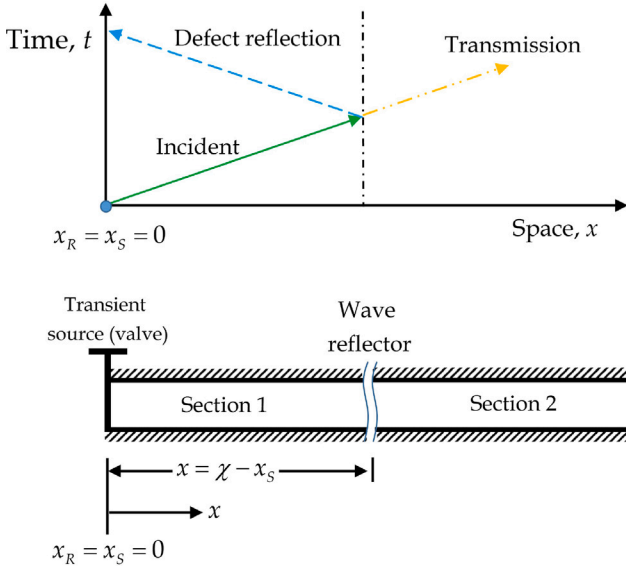


Fig. 1. Waveguide with a wave source (valve) at the end of the pipe and a single wave reflector located at a distance χ from the source location.

where $\mathcal{G}(t) = q_{in}(t) * G(t)$, $G(t)$ is given by Eq. (6), $\|\cdot\|$ represents the $\mathbb{L}_2$ -norm, and x_R is the location of sensor for transient response measurements. Using the Plancherel–Parseval theorem (Axisa, 2003, pp. 314), Eq. (9) is expressed in the frequency domain as

$$\hat{\chi} = \arg \max_{\tilde{\chi}} \tilde{B} = \arg \max_{\tilde{\chi}} \left\langle \frac{\tilde{\mathcal{G}}(\omega, x_R | \hat{\chi})}{\sqrt{\tilde{\mathcal{G}}^C(\omega, x_R | \hat{\chi}) \tilde{\mathcal{G}}(\omega, x_R | \hat{\chi})}}, \Delta \tilde{H}_m(\omega, x_R | \tilde{\chi}) \right\rangle \quad (10)$$

where $\langle \cdot, \cdot \rangle$ represents the inner product, $\tilde{\mathcal{G}}(\omega) = \mathcal{F}[\mathcal{G}(t)]$ and C in the superscript denotes the complex conjugate. Note that Eq. (10) is equivalent to the objective function of matched-field processing (MFP; Wang and Ghidaoui, 2018b, Eq. (16)). This means that the time-reversed signal acts as a ‘matched-filter’ in the frequency domain whose frequency spectrum depends on the estimate of χ . This further implies that the two complex wave fields (i.e., $\tilde{\mathcal{G}}(\omega, \hat{\chi})$ and $\Delta \tilde{H}_m(\omega, \chi)$ in Eq. (10)) have an optimal phase match only when $\hat{\chi} = \chi$. The equivalence of the proposed technique and MFP is not surprising because both methods are inherently based on TR technique. However, the strength of the proposed method lies in detecting multiple defects of different kinds simultaneously (as shown below) and the flexibility to truncate the undesired transient signal which is not the case for MFP.

Once the defect location $\hat{\chi}$ is determined by using Eq. (9) (or Eq. (10)), $\hat{A}^\chi$ is solved for using the following relation (see Eq. (C.3) in Appendix C)

$$\hat{A}^\chi = \left(\int_{-\omega_{\max}}^{\omega_{\max}} \tilde{\mathcal{G}}^C(\omega, \hat{\chi}) \tilde{\mathcal{G}}(\omega, \hat{\chi}) d\omega \right)^{-1} \cdot \int_{-\omega_{\max}}^{\omega_{\max}} \tilde{\mathcal{G}}^C(\omega, \hat{\chi}) \Delta \tilde{H}_m(\omega, \chi, \hat{A}^\chi) d\omega. \quad (11)$$

where $\omega_{\max}$ is the maximum angular frequency that is injected into the system. Note that according to Eq. (3), if $\hat{A}^\chi$ has a negative sign, it implies that the reflector is a mass source (leak), otherwise $\hat{A}^\chi$ represents a momentum source (blockage).

For a mass source reflector, the defect size s_L^χ is estimated by substituting $\hat{A}^\chi$ and $\hat{\chi}$ in the Eq. (3) and leads to

$$\hat{s}_L^\chi = \frac{4\hat{A}^\chi (gA)^2}{a_0 (a_0 \tilde{q}_S - gA\hat{A}^\chi)} \sqrt{\frac{\hat{H}_{0,\chi}}{2g}} \quad (12)$$

where $\hat{H}_{0,\chi} = H_0(x_R) + \hat{\chi} R Q_0(x_S)/(2gA)$ is the estimated value of steady-state pressure head at the estimated reflector location. Likewise, the head loss coefficient $\hat{k}_{HL}^\chi$ of a momentum source reflector is estimated by substituting $\hat{A}^\chi$ and $\hat{\chi}$ in Eq. (3) which gives

$$\hat{k}_{HL}^\chi = \frac{2gA^2 a_0 \hat{A}^\chi}{Q_0 (a_0 \tilde{q}_S - gA\hat{A}^\chi)}. \quad (13)$$

where $\tilde{q}_S = 1 \text{ m}^3$ is kept in Eqs. (12) and (13) to conserve the units, and $Q_0(x_S)$ is the steady-state flow rate at the source location.

1.3. Solution for multiple defects

Consider a pipe system as shown in Fig. 2 where a mass point source is located at χ_1 and a momentum point source is placed at χ_2 . The solution for $\Delta \tilde{h}$ is derived in Appendix D and given by Eq. (D.26) as

$$\Delta \tilde{h}(\omega, x_R | \tilde{\chi}) = \tilde{A}_{\text{mass}}^{\chi_1} \tilde{G}(\omega, \chi_1) + \tilde{A}_{\text{mom}}^{\chi_2} \tilde{G}(\omega, \chi_2) \quad (14)$$

where $\tilde{\chi} = (\chi_1, \chi_2)$. Note that Eq. (14) is a linear combination of reflected waves from reflector 1 and reflector 2. Therefore, a general expression of $\Delta \tilde{h}$ for N_R number of reflectors can be written as

$$\Delta \tilde{h}(\omega, x_R | \tilde{\chi}) = \sum_{j=1}^{N_R} \tilde{A}^{\chi_j} \tilde{G}(\omega, \chi_j) \quad (15)$$

where $\tilde{\chi} = (\chi_1, \dots, \chi_{N_R})$, and $\Delta \tilde{h}(\omega, x_R | \tilde{\chi})$ is the impulse response due to N_R number of reflectors in the system. For the case of no damping, the inverse Fourier transform of Eq. (15) gives

$$\Delta h(t, x_R | \tilde{\chi}) = \sum_{j=1}^{N_R} A^{\chi_j} G(t, \chi_j) \quad (16)$$

where

$$G(t, \chi_j) = \delta \left[t - \frac{2(\chi_j - x_S)}{a_0} \right] \quad (17)$$

Note that the reflection arrival time of each reflector is given by its own transfer function $G(t)$ which is independent of other reflectors' locations, given that the defect-defect interaction is negligible (Wang and Ghidaoui, 2018a) in comparison to the primary reflection of individual defects. As such, for multiple defect localization problem, a single reflector model can be employed to estimate the locations of multiple defects in the system. In such manner, the objective function B (in Eq. (9)) would have multiple significant peaks. Mathematically, the objective function for multiple defects search using one defect model can be written as

$$\tilde{\chi} = \arg \max_{\tilde{\chi}} B = \arg \max_{\tilde{\chi}} \int_{-\infty}^{\infty} \frac{\mathcal{G}(\tau, x_R | \hat{\chi})}{\|\mathcal{G}(\tau, x_R | \hat{\chi})\|} \cdot \Delta H_m(\tau, x_R | \tilde{\chi}) d\tau \quad (18)$$

or, in the frequency domain,

$$\tilde{\chi} = \arg \max_{\tilde{\chi}} \tilde{B} = \arg \max_{\tilde{\chi}} \left\langle \frac{\tilde{\mathcal{G}}(\omega, x_R | \hat{\chi})}{\sqrt{\tilde{\mathcal{G}}^C(\omega, x_R | \hat{\chi}) \tilde{\mathcal{G}}(\omega, x_R | \hat{\chi})}}, \Delta \tilde{H}_m(\omega, x_R | \tilde{\chi}) \right\rangle \quad (19)$$

Finally, the procedure to apply the proposed TR-based defect detection method is described in the following Algorithm.

2. Numerical test cases

In this section, numerical examples of multiple defects in a single pipeline are discussed using a transient model based on method of characteristics (MOC) (Wylie et al., 1993). Consider a pipe of length $L = 200 \text{ m}$ with a valve (wave source) at $x = x_S = 0$, and a reservoir at $x = L$. The internal diameter of the pipe is $D = 79.2 \text{ mm}$. Other pipe properties are listed in Table 1. The steady-state flow rate and

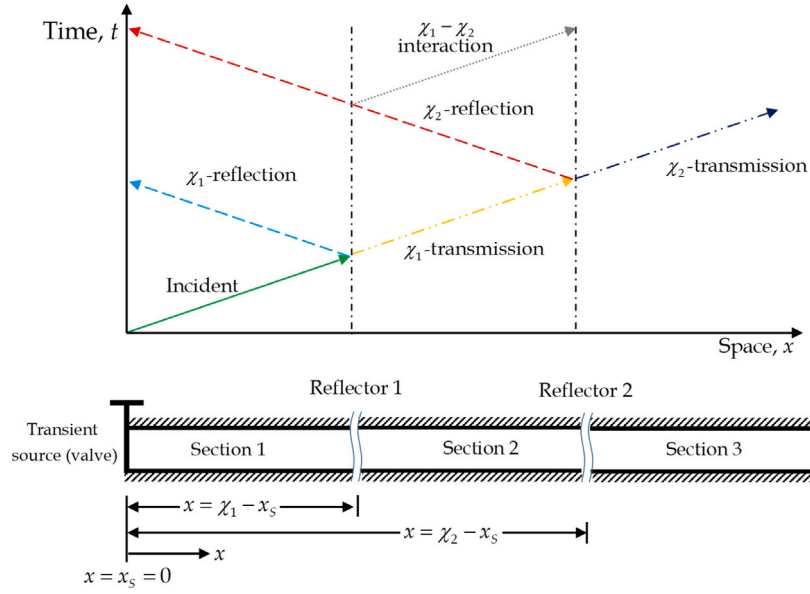


Fig. 2. Waveguide with two reflectors of different types.

Algorithm Defect detection procedure using TR technique

1. Excite the system from the point of access to the pipe system using a pulse-type forcing, record the pressure head response $H_m(t)$ and subtract the incident pulse from the measured response which is denoted by $\Delta H_m(t)$.
2. Estimate the wave speed a_0 from the data or using Eq. (A.3) and truncate the measured signal such that the length of the signal $t \in [0, 2l_k/a_0]$, where l_k denotes the length of the pipeline being inspected. For the case of system with unknown topology, l_k denotes the distance between measurement location x_R and location of the known boundary condition at the most upstream side.
3. Reverse (flip) the truncated signal in time such that $\Delta H_m(t)$ becomes $\Delta H_m(-t)$.
4. Determine $|B|$ using Eq. (9) (or Eq. (10) in frequency domain) for each $\hat{\chi}$ and plot $|B|$ with respect to $\hat{\chi}$. Each peak with a significant amplitude in the plot represents a potential defect candidate.
5. Determine the amplitude of the reflected wave $\hat{A}^\chi$ using Eq. (11) for each estimated defect location $\hat{\chi}$.
6. If $\hat{A}^\chi$ has a negative sign, $\hat{A}^\chi = \hat{A}_{\text{mass}}^\chi$ which is further characterized by its size s_L^χ that can be estimated using Eq. (12). Otherwise, $\hat{A}^\chi = \hat{A}_{\text{mom}}^\chi$ which is described by its head-loss coefficient k_{HL}^χ that can be estimated using Eq. (13).

pressure head at the valve are $Q_0 = 0.4$ l/s and $H_0 = 20$ m, respectively. Transient waves in the form of Gaussian pulses are generated by rapidly closing and opening the valve. The temporal width of the incident pulse is $T_p = 0.04$ s, where T_p denotes the valve maneuver time. The signals in these examples are collected at $x_R = x_S = 0$. Numerical simulations are performed for single and multiple (up to 3) defects. For each simulation, Gaussian white noise with zero mean and fixed variance is added to the signal. The noise variance is obtained by the following expression (Wang and Ghidaoui, 2018b; Waqar et al., 2019)

$$\sigma = |A^\chi| \times 10^{-\text{SNR}/20} \quad (20)$$

Table 1

System characteristics used for numerical simulations.

Variable	Description
Length, L	200 m
Internal diameter, D	79.2 mm
Wall thickness, e	5.4 mm
Elastic modulus, E	117 GPa (for elastic pipe) 1.47 GPa (for HDPE pipe)
Wave speed, a_0	1302 m s ⁻¹ (for elastic pipe) 324 m s ⁻¹ (for HDPE pipe)
Joukowski's pressure head, H_J	10.8 m (for elastic pipe) 2.7 m (for HDPE pipe)
Sampling frequency	2048 Hz
Minimum wavelength, $\lambda_{\min}$	26.04 m (for elastic pipe) 6.48 m (for HDPE pipe)
Steady friction factor, f_{DW}	0.02
Creep compliance, J	$[2.16, 0.73, 0.55, 0.25, 0.008] \times 10^{-10}$ Pa ⁻¹
Retardation time, τ	$[0.052, 0.62, 0.81, 6.56, 9.05]$ s

where σ is the standard deviation of noise, SNR is in decibels (dB) and $|A^\chi|$ is the absolute magnitude of reflected wave from the defect. For the case of multiple defects, $\min(|A^{\chi_1}|, |A^{\chi_2}|, \dots)$ is used in Eq. (20) to determine σ . The localization accuracy is determined by taking the absolute difference between the actual χ and the estimated $\hat{\chi}$. The diffraction limit (Waqar et al., 2022; Fink, 2006) is determined as $0.5\lambda_{\min} = a_0/(2f_{\max})$, where a_0 is the wave speed and $f_{\max}$ is the maximum frequency of the injected signal. If $|\chi - \hat{\chi}| \leq 0.5\lambda_{\min}$, the localization is deemed successful and unsuccessful otherwise. The numerical simulations are performed for both elastic and VE pipe. The VE effects are incorporated using Kelvin–Voigt model (Covas et al., 2005) and the unsteady friction is incorporated using Vardy–Brown model (Vardy and Brown, 1995)

2.1. Example 1: one leak case

A leak of size $s_L = 4.92$ mm² (i.e. 0.1% of pipe area, steady-state leak flowrate, $Q_{0,L} = 0.1$ l/s) is simulated at $\chi/L = 0.6$ from the valve. The simulated signals in the elastic pipe and in the VE pipe are shown in Fig. 3a and 3b, respectively. The VE model parameters (e.g., creep compliance and retardation time) are provided in Table 1. In Fig. 3, the horizontal axes are normalized with respect to L/a_0 (i.e. the characteristic time period of the system). The vertical axes are normalized with respect to the Joukowski pressure head H_J (Wylie

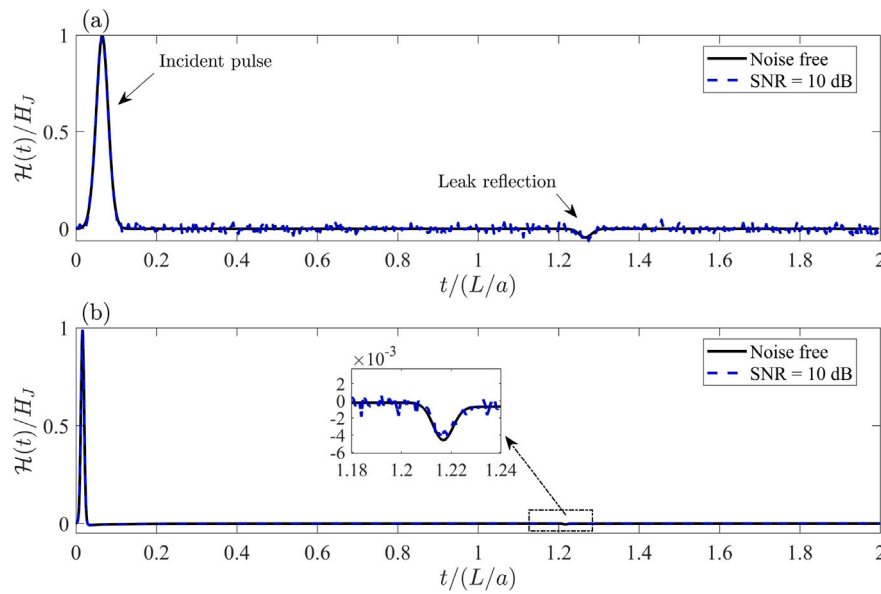


Fig. 3. Numerically simulated signals in (a) elastic pipe, and (b) visco-elastic pipe for one leak case.

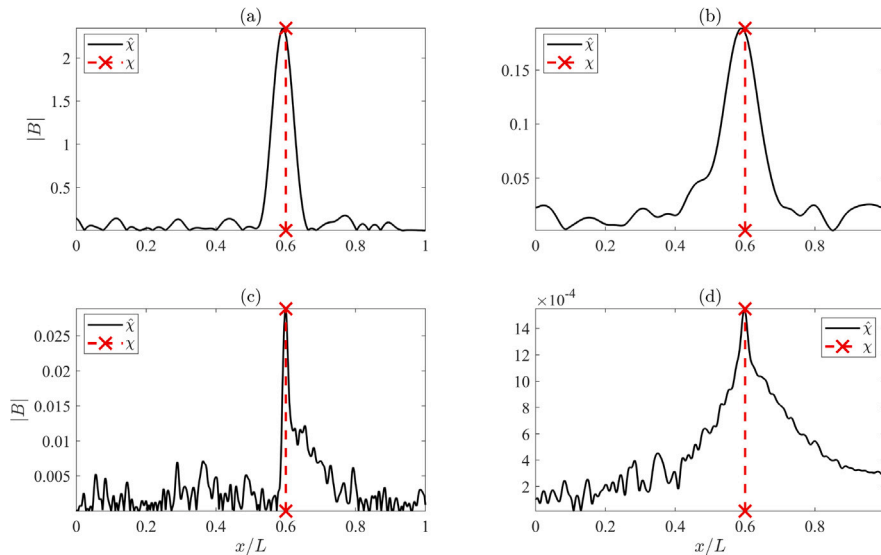


Fig. 4. Leak localization results for elastic pipe system where damping is (a) neglected and (b) included, and VE pipe system where frictional and VE effects are (c) neglected and (d) included. Dashed line shows the exact leak location.

et al., 1993). The difference in the temporal width of the incident pulse in elastic and visco-elastic pipe is due to the difference in the wave speed in the elastic and VE pipe systems (see the numeric values of wave speed in Table 1). Moreover, each plot in Fig. 3 contains a noise-free signal (solid line, for reference only) and a noisy signal (dashed line) with SNR = 10 dB, wherein, $\sigma \approx 0.3|A^x|$.

For defect localization, a vector for defect candidates is $\vec{\chi} = (0, 0.01, \dots, L)$. Following the Algorithm (step 1), the incident pulse (i.e., the pulse between $t \in [0, T_p]$, where $T_p = 0.04$ s) is subtracted from the signals to obtain $\Delta H(t)$. The obtained signal is then used for defect localization. The results are shown in Fig. 4a and 4b for elastic pipe and Figs. 4c and 4d for visco-elastic pipe, respectively. Damping is neglected in Fig. 4(a,c) but included in Fig. 4(b,d). In all subplots of Fig. 4, the solid lines represent the estimated defect location, dashed lines show the exact defect location. It can be seen from Fig. 4 that the models with and without damping accurately localize the defect with the average absolute localization error of 0.15 m. These results are supported by the findings in Waqar et al. (2022) where the

effect of damping was investigated in detail. While damping has little influence on leak localization (Fig. 4(a,c)), it has profound influence on determining the leak size. Therefore, only the model that include damping is used in the estimation of A^x . Following the Algorithm (step 6), the leak size is then estimated using Eq. (12) and found to be $\hat{s}_L = 4.94$ mm². The relative error between s_L and $\hat{s}_L$ is about 0.4% for both elastic and VE pipe.

2.2. Example 2: multiple leaks and spatial resolution

For multiple leaks case, two sub-cases are considered. Sub-case 1 includes two leaks located at $\chi_1/L = 0.3$ and $\chi_2/L = 0.6$. Their corresponding effective leak sizes are $s_{L1} = 4.92$ mm² (0.1% of pipe area; $Q_{0,L} = 0.1$ l/s) and $s_{L2} = 9.85$ mm² (0.2% of pipe area; $Q_{0,L} = 0.2$ l/s). Sub-case 2 includes an additional leak to sub-case 1 at $\chi_3/L = 0.8$ with the effective area of $s_{L3} = 7.38$ mm² (0.15% of pipe area; $Q_{0,L} = 0.15$ l/s). The numerical simulations are performed with the same system variables as defined in previous sub-section (or see Table 1) and the

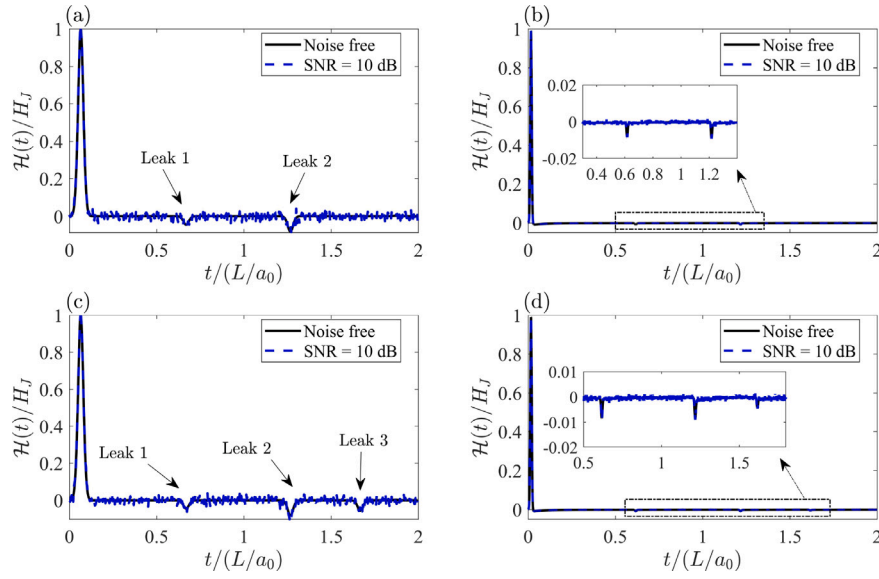


Fig. 5. Numerically simulated signals with two leaks in (a) elastic pipe & (b) VE pipe, and with three leaks in (c) elastic pipe & (d) VE pipe.

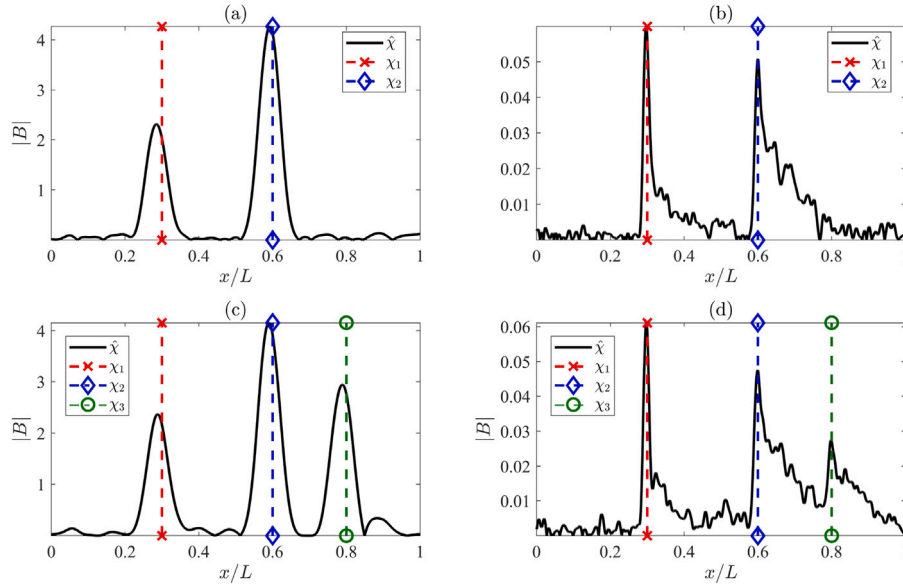


Fig. 6. Results for two leaks case in (a) elastic pipe & (b) VE pipe, and for three leaks case in (c) elastic pipe & (d) VE pipe. Vertical lines show the exact leak locations.

acquired signals are plotted in Fig. 5. Figs. 5a and 5b corresponds to two leaks case in the elastic and the VE pipe, respectively. Figs. 5c and 5d correspond to three leaks case in the elastic and VE pipe, respectively.

For the two leaks case, the localization results are presented in Figs. 6a (for elastic pipe) and 6b (for VE pipe). For three leaks case, the results are given in Figs. 6c and 6d for elastic and VE pipes, respectively. It can be seen that all the leaks are accurately localized in all scenarios. Note that the search for multiple leaks is conducted using the one leak model as discussed in Section 1.3. The absolute localization error for the two and three leaks cases is smaller than $0.5\lambda_{\min}$. In terms of leak size estimate, the results are presented in Fig. 7 wherein the real leak sizes are denoted by solid bars and the estimated leak sizes are shown by the vertical lines. In all scenarios, the absolute relative error among various estimated leaks' sizes ranges between 7% to 18%.

The accuracy of the proposed method in the context of detecting two closely located defects is also investigated. Let d_L be the distance between two consecutive leaks. Numerical simulations with variable d_L ranging between $0.20\lambda_{\min}$ to $0.65\lambda_{\min}$ are performed and the selective

leak localization results are presented in Fig. 8. All the results in Fig. 8(a to d) are for the case of elastic pipe. It can be seen that the proposed TR technique successfully localizes the leaks that are as close as $0.35\lambda_{\min}$. Same findings are also applicable for visco-elastic pipe system; and therefore, the results are not presented here for the sake of brevity. Moreover, for higher resolution, high-frequency acoustic waves can be used which excite radial wave modes and provide B- or C-scans of the pipe (Louati and Ghidaoui, 2017a,b; Nasraoui et al., 2023).

2.3. Example 3: mixed-type of defects

In this example, two scenarios are considered: In scenario I, one leak and one discrete blockage are simulated at $\chi_1/L = 0.4$ and $\chi_2/L = 0.6$, respectively. The effective leak size is $s_L = 4.92 \text{ mm}^2$ (0.1% of pipe area; $Q_{0,L} = 0.1 \text{ l/s}$). The pressure head loss coefficient of the discrete blockage is $k_{HL}^{\chi_2} = 1000$ which corresponds to 0.62 m of pressure head loss in the steady-state condition. Other system properties are same as previous example. The simulated transient signals are presented in

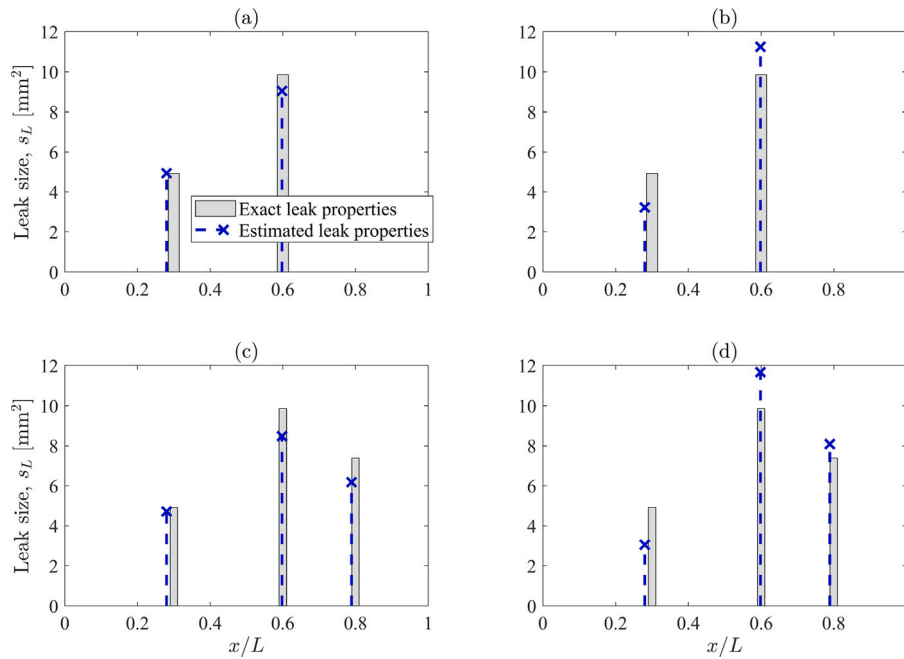


Fig. 7. Estimated leak sizes for two leaks case in (a) elastic pipe & (b) visco-elastic pipe, and for three leaks case in (c) elastic pipe & (d) visco-elastic pipe. In this plot, the location of bar (or line) indicate the defect location and the height of bar (or line) shows the size of the defect.

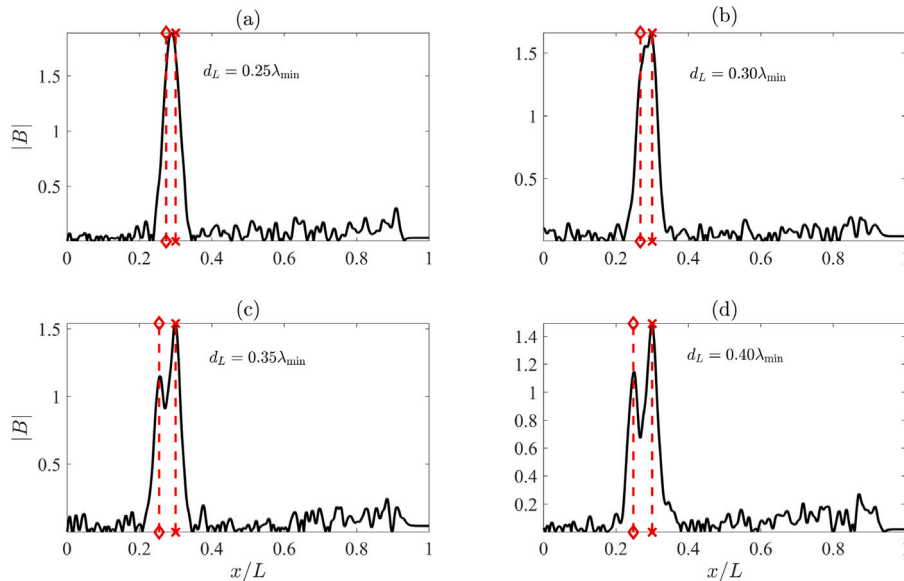


Fig. 8. Spatial resolution of two closely placed leaks with distance $d_L =$ (a) $0.25\lambda_{\min}$, (b) $0.30\lambda_{\min}$, (c) $0.35\lambda_{\min}$, and (d) $0.40\lambda_{\min}$.

Fig. 9a (for elastic pipe) and in Fig. 9b (for VE pipe). In scenario II, an additional discrete blockage is simulated at $x_3/L = 0.8$ whose loss coefficient is $k_{HL}^{x_3} = 1200$ and produces 0.52 m of pressure drop during the steady-state condition. The corresponding numerical signals are presented in Figs. 9c (for elastic pipe) and 9d (for visco-elastic pipe).

Fig. 10 shows the defect localization results for both scenario I (10a and 10b) and scenario II (10c and 10d). In all cases, the proposed method has successfully localized the defects as the exact defect locations and multiple peaks of the objective function $|B|$ are within the interval of $0.5\lambda_{\min}$ (see Table 1). The overall absolute defect localization error is less than 0.2 m. After localization, the next step is to identify the type of the defect and its corresponding size. This task is performed according to step 6 of the Algorithm and the results are shown in Fig. 11. In particular, Figs. 11a and 11c show the estimated parameters

for scenario I wherein both leak and discrete blockage are successfully classified for both elastic and VE pipes. For scenario II, where there is one leak and two discrete blockages, the results are presented in Figs. 11b and 11d, respectively. In terms of leak size $\hat{s}_L$ and head loss coefficient estimation $\hat{k}_{HL}^x$, the overall relative error is between 10%–25%. The higher value of relative error is ascribed to the fact that single defect model is used to match the data with two defects. This error can be minimized using multi-defect model wherein the search space is multi-dimensional (instead of one dimensional). More details on this approach can be found in Wang and Ghidaoui (2018a).

3. Stability of the defect detection scheme

In this section, the stability of the proposed method is investigated with respect to the SNR and uncertain wave speed.

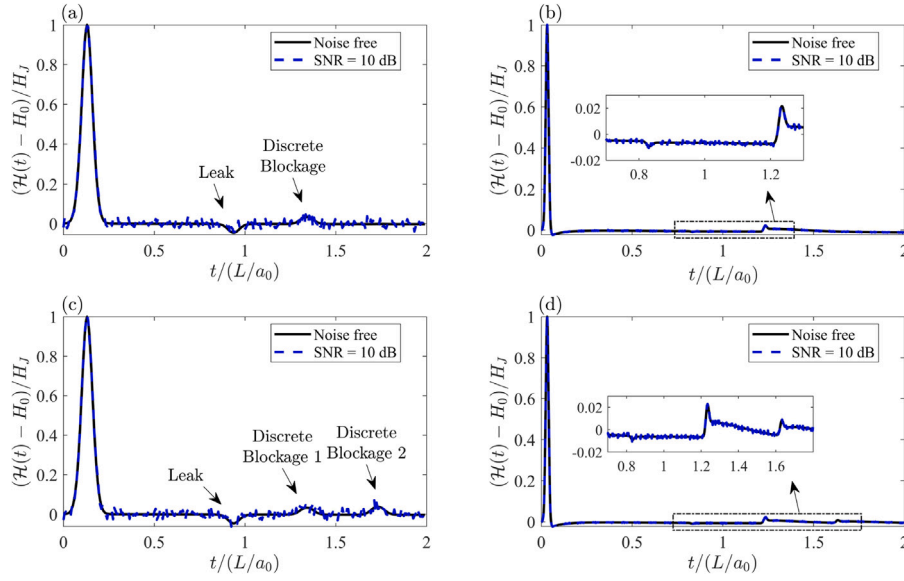


Fig. 9. Numerically simulated signals with two different defects in (a) elastic pipe & (b) VE pipe, and with three defects of mixed-type in (c) elastic pipe & (d) VE pipe.

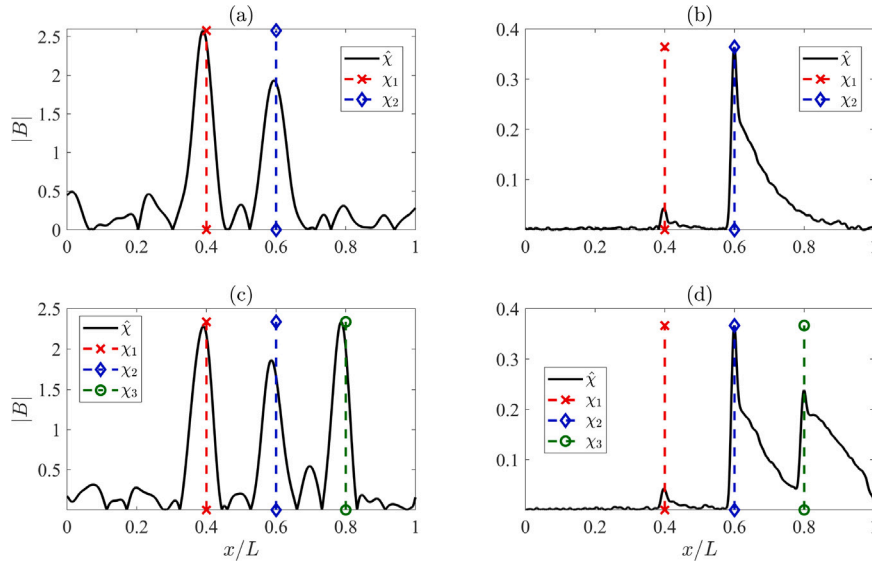


Fig. 10. Results for two defects of different types in (a) elastic pipe & (b) VE pipe, and for three defects of different types in (c) elastic pipe & (d) VE pipe. Vertical lines show the exact defect locations.

3.1. SNR

The influence of different values of SNR ranging between 10 dB ($\sigma \approx 0.3|A^x|$) to -5 dB ($\sigma \approx 1.8|A^x|$) on the leak localization and leak size estimation is shown in Fig. 12a and 12b, respectively. In Fig. 12, solid lines represent the elastic pipe system and dashed lines represent the VE pipe system. Each line shows the average value of 30 simulations and vertical bars show the 95% confidence. It can be seen that for $\text{SNR} \geq 0$ dB, the absolute leak localization error is $< 0.5\lambda_{\min}$ and the absolute leak size error is $\leq 1 \text{ mm}^2$ for both the elastic and the visco-elastic pipes. However, for $\text{SNR} < 0$ dB, leak localization accuracy is relatively more affected for the visco-elastic pipe than the elastic pipe. It is because the reflection coefficient of a leak in the visco-elastic pipe is relatively smaller than that in the elastic pipe. Therefore, for a given SNR, it is more difficult to detect a small leak in visco-elastic pipe than in elastic pipe for a given incident wave. One possible treatment to detect such small leaks under low SNR would be to conduct multiple transient tests and obtain the average signal (Wang et al., 2020b).

3.2. Uncertain wave speed

In real-field experiments, the theoretical and experimental wave speeds, denoted as $\hat{a}_{th}$ and a_{exp} , respectively, often has discrepancy which may goes up to 15% (Wylie et al., 1993; Wang and Ghidaoui, 2018b). In terms of defect localization, imprecise estimate of wave speed leads to inaccurate results (Wang et al., 2020a). For this purpose, accuracy of the proposed method is tested under uncertain wave speed whereby $\hat{a}_{th}$ is varied up to $\pm 15\%$. The numerical simulations are performed for one leak case with $\text{SNR} = 10$ dB. The results are presented in Figs. 13a and 13b for elastic and VE pipe systems, respectively. In both figures, the x-axes show the non-dimensional wave speed variability and y-axes show the normalized absolute error in the estimated leak location. The region below 0.5 on the vertical scale represent the reliable localization as the error between the estimated and exact leak location in this region is smaller than $0.5\lambda_{\min}$. In both pipe system, leaks are localized using both damping-free and damped wave model and their corresponding results are found to be consistent

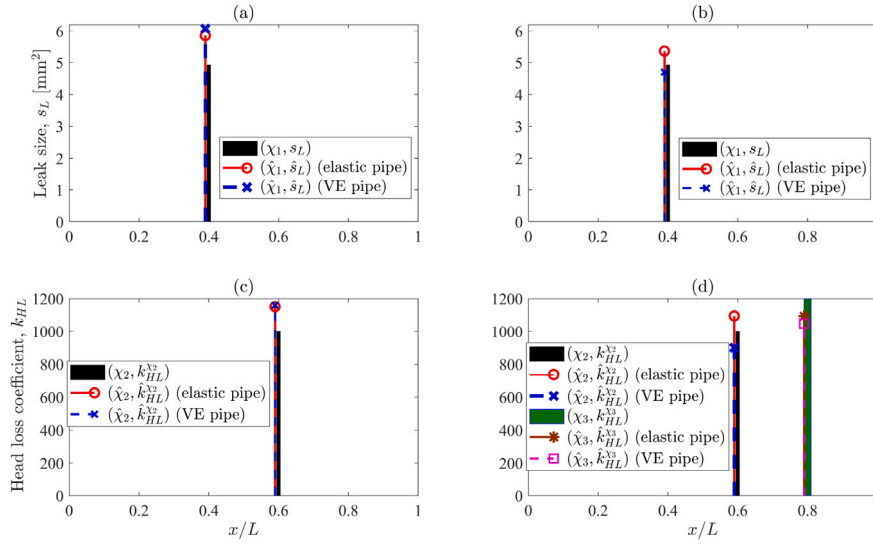


Fig. 11. Estimated leak size and head loss coefficient for scenario I in (a) & (c), and for (ii) scenario II in (b) & (d) respectively. Each subplot contains exact defect parameters, estimated defect parameters in elastic pipe and estimated defect parameters in VE pipe.

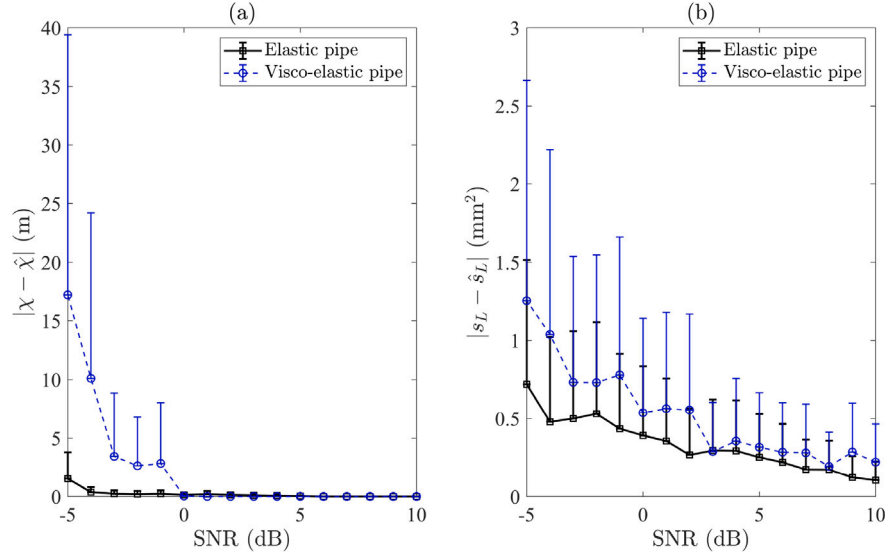


Fig. 12. Absolute error in single (a) leak localization, and (b) leak size estimation with respect to SNR.

with each other. These plots indicate that, for elastic pipe (Fig. 13a), the interval between exact and estimated defect is relatively large which covers about $\pm 10\%$ uncertainty in wave speed and still gives the localization error smaller than $0.5\lambda_{\min}$. For the visco-elastic pipe (Fig. 13b) only 2.5% of uncertainty in wave speed gives the localization error within the half wavelength interval. This implies that if $\hat{a}_{th}$ varies more than 2.5% of the experimental wave speed in visco-elastic pipe, the localization error would be larger than the half-wavelength interval. It is because the wavelength in visco-elastic pipes is much smaller than the wavelength in the elastic pipe; and therefore, the tolerance in wave speed uncertainty in visco-elastic pipe is smaller than that for the elastic pipe system.

To further illustrate the foregoing analysis, the objective function plots with 10% variability in the wave speed in an elastic and 2.5% variability in VE pipe systems are given in Figs. 13c and 13d, respectively. In these figures, the x-axes are centered with respect to actual leak location χ and normalized with respect to minimum wavelength

and y-axes show the normalized objective function $|B|$, the solid line represents the case where $\hat{a}_{th} = a_{exp}$, and the dash line denotes the case when the uncertain value of wave speed is used. In both Figs. 13c and 13d, the position of the peak of dash line coincides with the first zero of the solid line, implying that the error between estimated and exact leak location would exceed the half-wavelength interval if the variation in the theoretical wave speed is further increased. Overall, these results show that the proposed technique can handle uncertain wave speed up to some level depending on the pipe properties.

4. Experimental validation

In this section, the proposed TR technique is validated through laboratory experiments. The test rig is comprised of a Pump-Pipe-Valve system and situated at the Water Research Resources Laboratory of the Hong Kong University of Science and Technology (HKUST). The schematic of the system is given in Fig. 14. A buffer tank (BT) is located

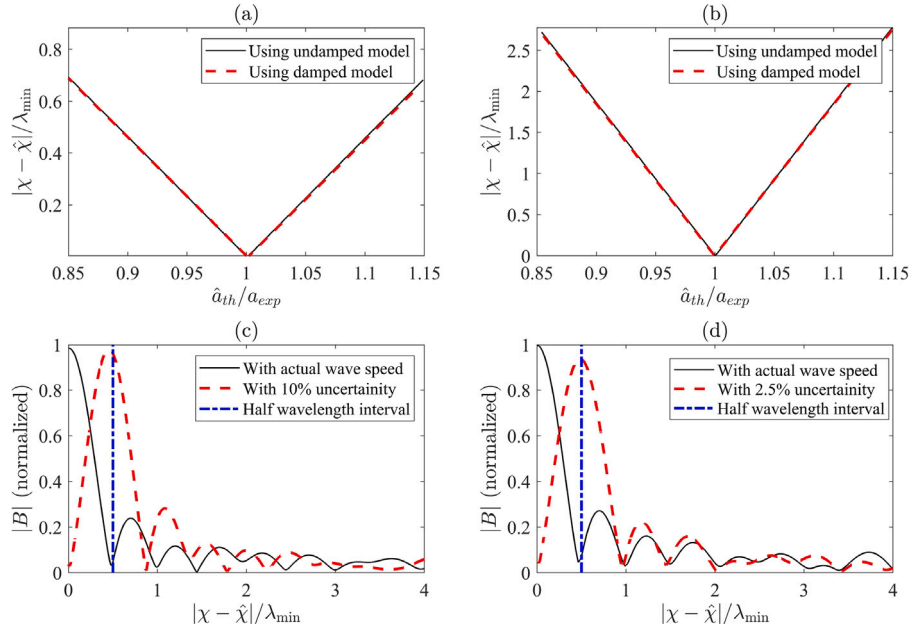


Fig. 13. Sensitivity analysis for leak localization with uncertain wave speed using undamped model (solid line) and damped model (dash line) in (a) elastic pipe, (b) visco-elastic pipe. Objective function plots with exact wave speed (solid line) and with uncertain wave speed (dash line) in (c) elastic pipe, and (d) visco-elastic pipe.

at the extreme upstream side that supplies water through a booster pump (BP) (Model: KIKAWA KRS16-3) to the 145.4 m long high density polyethylene (HDPE-PE100) pipe. The internal diameter of the pipe is 79.2 mm. Other pipe properties including the VE modeling parameters (i.e., creep compliance coefficients and retardation time) are provided in Table 2. On the downstream end, the pipe is connected back to BT resulting in a recirculating system. A gate valve (GV7) is installed between downstream end of pipe and BT to regulate the flow in the system. Along the pipe, there are fourteen manual ball valves of small diameter (12.7 mm) and six additional GVs installed. These small ball valves are referred as manual valves (MVs) which can be used to release the entrapped-air in the system, to install pressure transducers or to simulate leaks in the system. Due to the limited space in the laboratory, combination of straight pipe sections, curved pipe sections, elbows and tees are used in the setup. The degree of curvatures varies from 70° to 90° and the radius of curvatures varies from 20 cm to 30 cm depending on the setup orientation. This experimental facility also includes (i) an air bladder tank (ABT) as an upstream boundary condition, (ii) a 20 m long branch pipeline, (iii) a discrete blockage, and (iv) a portable pressure wave maker (PPWM) at the downstream end. In this paper, ABT, PPWM, and branch line are not included (i.e., isolated by valves). Moreover, GV7 is fully closed and the flow is regulated through a 12.7 mm diameter valve located at MV1. Thus, the effective pipe length from MV1 (downstream) to BP (upstream) is $L = 142.08$.

4.1. Defect simulation, transient generation and data acquisition

The steady-state flow rate is established through MV1 (fully open). The transient wave is generated by manually and fully closing MV1 in time T_c ranging between 0.025 s to 0.043 s. The transient response of the system is recorded at $x_R = MV2 = 2.46$ m using a UNIK5000 pressure transducer with the range of 0.5 MPa (5 bar) and with the accuracy of 0.04% (full scale best straight line) as shown in Fig. 14. Another pressure transducer of the same specifications is also placed at $x = MV4 = 22.65$ m to determine the wave arrival time for wave speed estimation. The data is collected at a sampling frequency of 1000 Hz using the National Instrument cRIO-9030 data acquisition system. In addition, two ultrasonic flow meters (Model: DXNP-UHS-NN) are also installed near the upstream and downstream ends to estimate

Table 2
Test rig pipe properties.

Property	Value
Internal diameter, D	79.2 mm
Wall thickness, e	5.4 mm
Elastic modulus, E	1.66 GPa
Friction factor, f_{DW}	0.028
Poisson's ratio, ν	0.43
Creep compliance, J	$[1.206, 1.417, 0.902, 0.558, 0.096] \times 10^{-10}$ Pa $^{-1}$
Retardation time, τ	$[0.0570, 0.5644, 0.6355, 3.9655, 10.9254]$ s

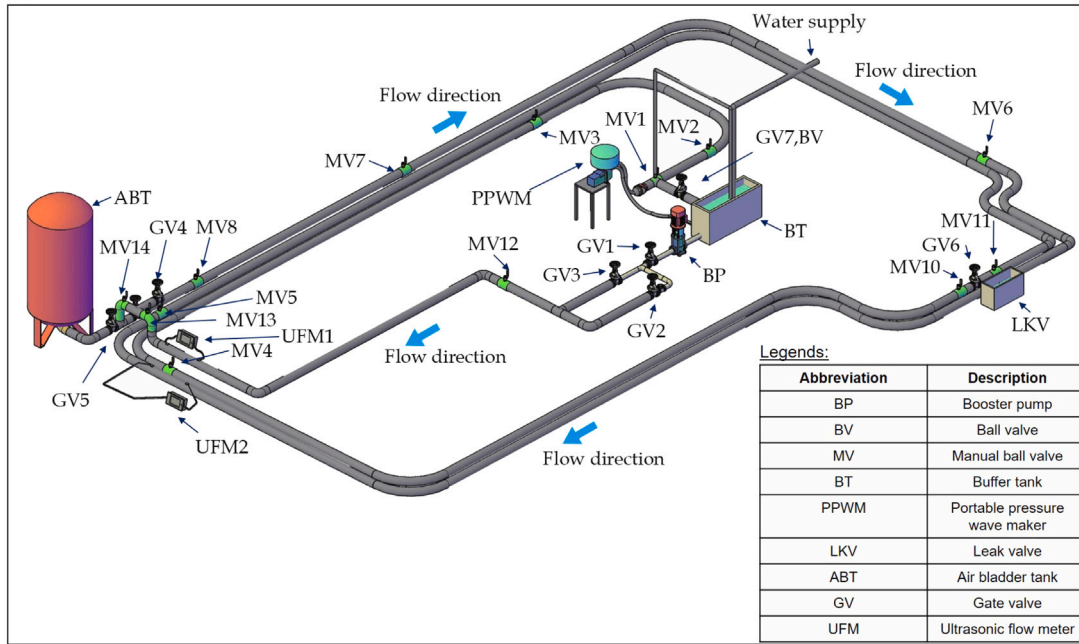
the steady-state flow rate in the system. Leaks are simulated using a side-discharge valve located at LKV. Discrete blockage is simulated using an in-line gate valve located at GV6.

4.2. Results and discussion

Six cases of one leak with variable leak sizes, three cases of one discrete blockage with variable sizes, and three cases of two defects (a leak and a discrete blockage) are considered. The detailed description of these test cases (including the estimated wave speed, valve time closure, steady-state flowrate and pressure head at the valve) is provided in Table 3 where the test cases are sorted according to the measured SNR. The measured pressure head response of the system in each test case is given in Fig. 15. For each case, the step response is transformed into a pulse response by creating a copy of the original step signal with a delay of about T_c seconds and subtracting the delayed signal from the original step response as suggested by Lee (2005). A low pass filter is then applied to minimize electromagnetic interference induced by the pump and other electrical appliances located near the data acquisition device. Only the first half-period of the signal is used for defect detection as it contains the sufficient information to localize defects in the system (Covas and Ramos, 2010; Ferrante et al., 2009; Meniconi et al., 2011; Zouari et al., 2020).

For defect detection, the average wave speed of $\bar{a}_0 = 357.3$ m s $^{-1}$ is used. The estimated defect location for each test case is presented in Fig. 16. The results show that for SNR ≥ 0 dB (i.e., C1 to C10), the defects are accurately localized. The average absolute defect localization error is around 3.2 m and is within the interval of $0.5\lambda_{\min} = 0.5\bar{a}_0T_c =$

a) 3D schematic



b) 2D layout

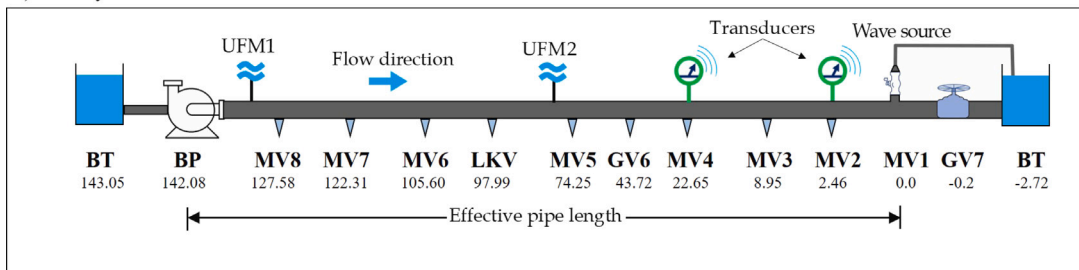


Fig. 14. Experimental setup at Water Research Resources Laboratory of the Hong Kong University of Science and Technology, Hong Kong.

Table 3

System properties during different test cases. The leak and discrete blockage are located at $x_L = 97.9$ m and $x_{DB} = 43.72$ m, respectively from the downstream valve location.

Case	Q_0 (l s ⁻¹)	H_0 (m)	T_c (s)	a_0 (m s ⁻¹)	Defect type	$Q_{L,0}$ (l s ⁻¹)	s_L (mm ²)	k_{HL}^x	SNR (dB)
C1	0.487	20.82	0.026	356.46	Leak	2.89	142.86	–	9.68
C2	0.307	23.77	0.040	358.49	Leak + DB	2.13	98.53	3000	8.00
C3	0.318	25.26	0.025	357.15	Leak + DB	1.65	73.89	3000	7.74
C4	0.577	27.03	0.043	359.25	Leak	3.06	133.01	–	6.17
C5	0.543	21.72	0.025	358.21	DB	–	–	775.89	6.15
C6	0.323	25.43	0.026	354.21	Leak + DB	1.65	73.89	2200	5.81
C7	0.324	25.67	0.028	354.69	DB	–	–	2200	4.85
C8	0.519	21.77	0.024	355.69	DB	–	–	518.72	4.69
C9	0.582	26.89	0.039	359.26	Leak	2.26	98.53	–	2.38
C10	0.487	26.59	0.028	362.52	Leak	3.60	157.64	–	0.26
C11	0.547	21.13	0.030	356.46	Leak	1.81	88.67	–	–2.03
C12	0.530	22.09	0.025	358.86	Leak	1.13	54.19	–	–6.26

6.98 m, where $T_c = 0.024$ s is considered as it is the fastest valve closure among all test cases. For the cases where SNR < 0 dB (i.e. C11, C12), the peak of the objective function $|B|$ is far from the exact defect location, implying inaccurate localization. It is because, in these cases, the reflection signatures of the defects are completely buried under the electromagnetic interference and hydraulic noise of the system. These outcomes are previously predicted in Fig. 12 (for VE pipe) where the accuracy of the proposed method is tested under various noise levels. In terms of defect type identification and its corresponding size estimation, the results are presented in Figs. 17 (for leaks) and 18 (for discrete blockages). The maximum relative error in the estimated

size is around 19.6% (for leak cases) and 20.2% (for discrete blockage cases).

5. Conclusions

In this paper, the problem of defect detection in pressurized fluid-filled pipelines is addressed. A time-reversal technique is proposed for discrete defect detection of mixed-type such as leaks and discrete blockages in the elastic and the visco-elastic pipe systems. The solution for defect localization is based on time-reversal technique, and the solution for defect type and its corresponding size estimate is obtained

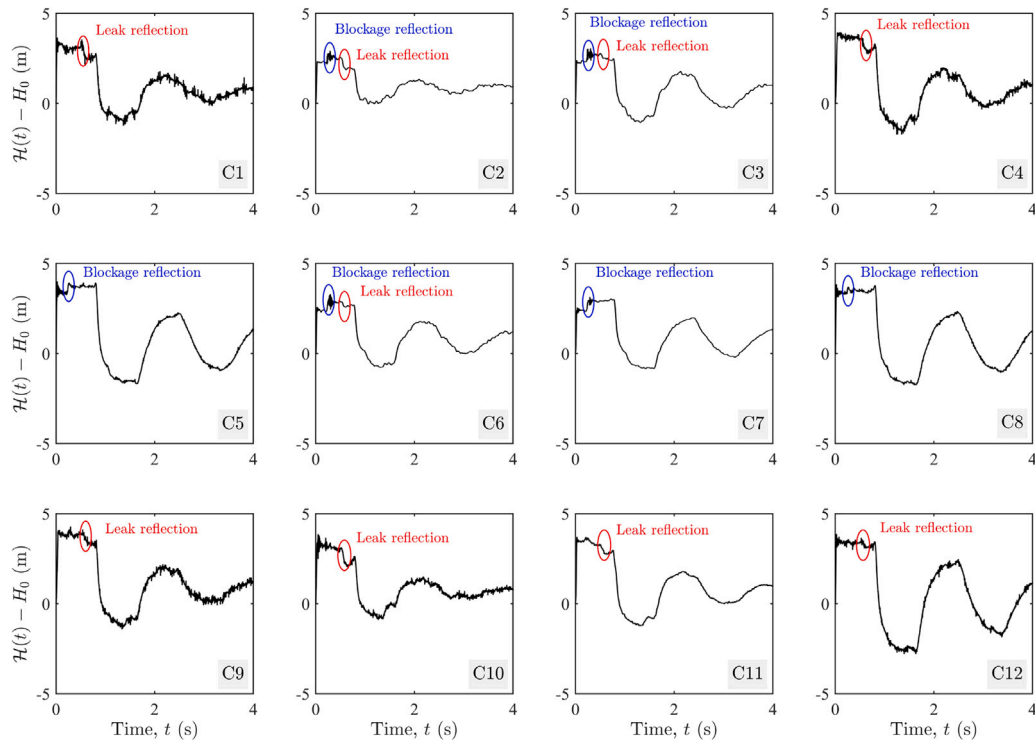


Fig. 15. Measured pressure head signals at MV2 for single defect (C1, C4, C5, C7-C12) and multiple defects (C2, C3, C6) defect.

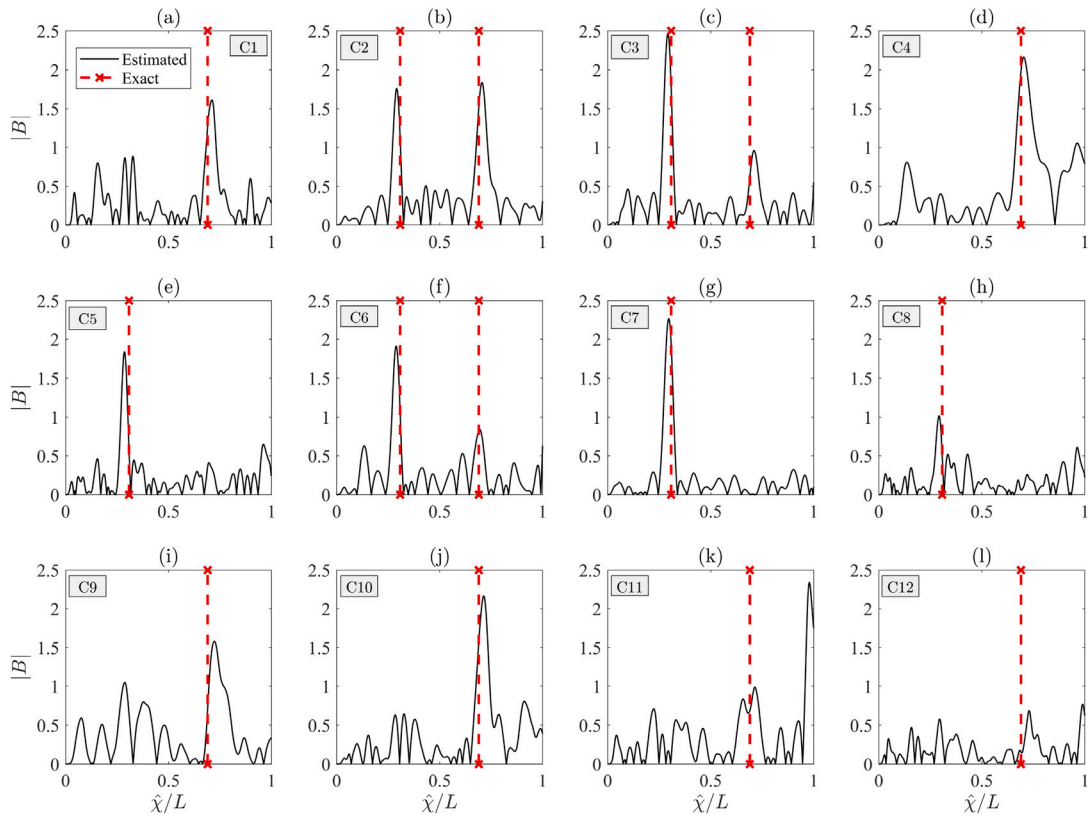


Fig. 16. Defect localization results for all test cases. Vertical dashed-lines show the exact defect location.

using the maximum likelihood estimation procedure. Moreover, the proposed TR technique is formulated to consider only the first half-period of the transient signal. The proposed technique is proved to be optimal in the sense of signal to noise ratio provided that the noise

distribution is normal and can be applied in both the time and the frequency domains.

Numerical simulations for single leak, multiple leaks, and mixed-type of defects (e.g., leak and discrete blockage) in both an elastic and

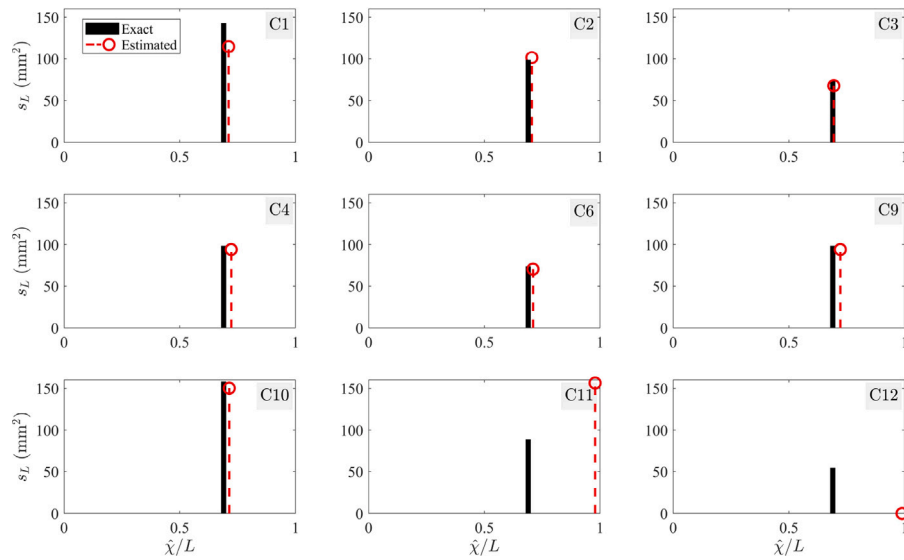


Fig. 17. Estimated leak location and leak size for Case C1–C4, C6, C9–C12. Solid bar represents the exact leak properties and dash line denotes the estimated leak properties.

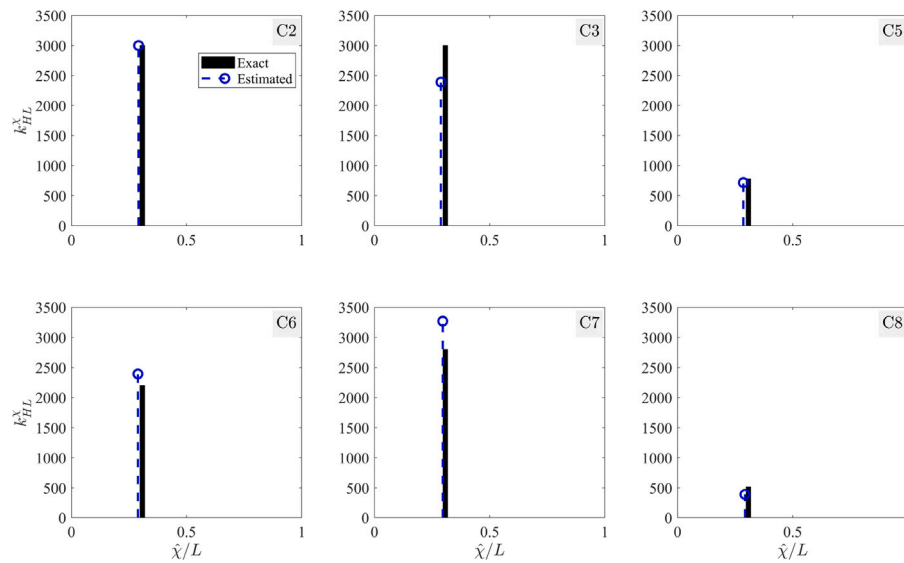


Fig. 18. Estimated discrete blockage location and head-loss coefficient for C2, C3, C5–C8. Solid bar represents the exact blockage properties and dash line denotes the estimated blockage properties.

a visco-elastic pipes are considered. In all cases, the proposed technique has accurately localized and identified the defects with the absolute localization error smaller than the half wavelength. In addition, it is demonstrated that an undamped wave model can be used to localize defects in damped systems such as the elastic pipe which has steady as well as unsteady friction or visco-elastic pipe which includes steady friction, unsteady friction and visco-elastic effects. In terms of resolution, the proposed technique has successfully localized two consecutive defects which are 0.35–wavelength distance apart in both elastic and visco-elastic pipes. Moreover, the stability of the proposed method is investigated under various signal-to-noise ratios and uncertain wave speed. For stability against noise, the SNR is varied between 10 dB to -5 dB. The results show that the localization error in elastic pipe remains within the interval of half wavelength for $\text{SNR} \geq -5$ dB. For visco-elastic pipe, however, the localization error is relatively larger for $\text{SNR} < 0$ dB. For the case of uncertain wave speed, the results show that the localization error remains within the half wavelength interval for up to 10% uncertainty in wave speed. For visco-elastic pipe, however, only 2.5% uncertainty in wave speed gives the reliable localization.

The proposed method is also validated with laboratory experiments. In particular, six cases of single leak with variable size, three cases of single discrete blockage with different head-loss coefficient and three cases of two defects (a leak and a blockage) are considered. The results show that for cases with $\text{SNR} \geq 0$ dB, the proposed method is quite accurate in terms of localization, defect type determination and defect size estimation. For $\text{SNR} < 0$ dB, on the other hand, the accuracy of the method is affected by the noise in the system. This issue can be addressed by conducting multiple measurements and taking the average pressure head response. Overall, the proposed TR technique has shown excellent results to be considered for further development.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

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Appendix A. Detailed derivation of analytical models

To describe the transient response of the system, the one dimensional water-hammer equations (i.e., mass and momentum conservations) are used. These two equations in the frequency domain are written as

$$\frac{\partial \tilde{q}}{\partial x} + \frac{gA}{a_{ve}^2} i\omega \tilde{h} = 0 \quad (\text{A.1a})$$

$$\frac{\partial \tilde{h}}{\partial x} + \frac{1}{gA} (i\omega + R + i\omega \hat{R}) \tilde{q} = 0, \quad (\text{A.1b})$$

where x is the spatial coordinate, $\tilde{h}$ is the pressure head perturbation, $\tilde{q}$ is the flow rate perturbation, g is the gravitational constant, A is the cross-sectional area of the pipe, $i = \sqrt{-1}$,

$$a_{ve} = \left[\frac{1}{a_0^2} + \frac{\rho D}{e} (1 - \nu^2) \sum_{j=1}^{N_{kv}} \frac{J_j}{1 + i\omega \tau_j} \right]^{-1/2}, \quad a_{ve} \in \mathbb{C} \quad (\text{A.2})$$

is the frequency-dependent wave speed (Wang et al., 2019), wherein

$$a_0 = \left(\frac{\rho}{K} + (1 - \nu^2) \frac{\rho D}{eE} \right)^{-1/2}, \quad a_0 \in \mathbb{R}, \quad (\text{A.3})$$

is the elastic wave speed, ρ is the density of the fluid, K is the bulk modulus of fluid, ν is the Poisson's ratio, E is the elastic modulus of pipe, D is the inner diameter of the pipe,

$$\hat{R}(\omega) = \frac{4}{D} \sqrt{\frac{\nu}{i\omega + \beta}}, \quad \hat{R} \in \mathbb{C}, \quad (\text{A.4})$$

is the frequency-dependent wall shear stress model to govern the unsteady-friction (Vardy and Brown, 1995), $\nu = 10^{-6} \text{ m}^2\text{s}^{-1}$ is the kinematic viscosity of fluid, $\kappa = \log(14.3/\text{Re}^{0.05})$, $\beta = 0.54\nu\text{Re}^\kappa/D^2$, $\text{Re} = Q_0 D/A\nu$ is the Reynolds number, R is the linearized steady-state friction term

$$R = (f_{DW} Q_0) / (DA),$$

f_{DW} is the Darcy–Weisbach friction factor, Q_0 is the steady-state flow rate, e is the pipe wall thickness, ω is the angular frequency, $\mathbb{R}$ denotes real variable, $\mathbb{C}$ denotes complex variable, J is the creep compliance, τ is the strain retardation time and N_{kv} represents the Kelvin-Voigt model elements. Differentiating Eq. (A.1a) with respect to x and subtracting it from Eq. (A.1b) gives the one-dimensional wave equation in terms of flow rate as

$$\frac{\partial^2 \tilde{q}}{\partial x^2} + \left(\frac{\omega^2}{a_{ve}^2} - \frac{i\omega}{a_{ve}^2} R + \frac{\omega^2}{a_{ve}^2} \hat{R} \right) \tilde{q} = 0 \quad (\text{A.5})$$

For the pipe system shown in Fig. 1, consider that the wave is injected through a mass point source in the form of $\delta(x - x_S, t - t_S)$, where x_S is the location of the source and t_S denotes the timing of the source. The source term can also be expressed as $\tilde{q}(\omega, x_S | x_S) = \tilde{q}_S(\omega) = 1$, where $\tilde{q}_S$ denotes the unit strength of the point source in the frequency domain. The general expression for excitation in terms of $\tilde{q}(x)$ is written as

$$\tilde{q}(\omega, x | x_S) = \tilde{q}_S \exp[-ik(x - x_S)] \quad (\text{A.6})$$

Taking the second derivative of Eq. (A.6) with respect to x and substituting in Eq. (A.5) gives

$$k = \sqrt{\frac{\omega^2}{a_{ve}^2} - \frac{i\omega}{a_{ve}^2} R + \frac{\omega^2}{a_{ve}^2} \hat{R}}; \quad k \in \mathbb{C} \quad (\text{A.7})$$

where k is known as the wave number. For pipe Section 1 in Fig. 1, the expression for incident and reflected wave from a reflector can be written as

$$\tilde{q}(\omega, x | x_S) = \tilde{q}_S \exp[-ik(x - x_S)] + \tilde{q}_R \exp[ik(x - x_S)] \quad (\text{A.8})$$

where, $\tilde{q}_R$ is the reflected wave from the reflector. Substituting Eq. (A.8) in Eq. (A.1a) and solving for $\tilde{h}(\omega, x | x_S)$ yields

$$\tilde{h}(\omega, x | x_S) = Z [\tilde{q}_S \exp[-ik(x - x_S)] - \tilde{q}_R \exp[ik(x - x_S)]] \quad (\text{A.9})$$

where $x \in [x_S, \chi]$, χ is the reflector location, and

$$Z = \frac{a_{ve}}{gA} \sqrt{1 - \frac{i}{\omega} R + \hat{R}}; \quad Z \in \mathbb{C} \quad (\text{A.10})$$

is the characteristic impedance of the pipe. Similarly, the expression for flow rate perturbation $\tilde{q}$ and pressure head $\tilde{h}$ in Section 2 (see Fig. 1) are defined as

$$\tilde{q}(\omega, x | x_S) = \tilde{q}_T \exp[-ik(x - x_S)], \quad (\text{A.11a})$$

and

$$\tilde{h}(\omega, x | x_S) = Z \tilde{q}_T \exp[-ik(x - x_S)], \quad (\text{A.11b})$$

respectively, where $x \geq \chi$ and $\tilde{q}_T$ is the amplitude of the transmitted wave through the reflector. To further develop the model, two kinds of reflectors are considered: (i) mass source reflector (for example, leak, active unauthorized demand, etc.), and (ii) momentum source reflector (discrete blockage, partially closed in-line valve, etc.).

A.1. Scenario I: Mass source reflector

Consider that the reflector is a mass point source which extracts fluid at position χ at a rate $\tilde{q}_\chi$ from the main pipe. The pressure head and flow conditions at such type of reflector are

$$\lim_{\varepsilon \rightarrow 0} \tilde{h}(\chi - \varepsilon) = \lim_{\varepsilon \rightarrow 0} \tilde{h}(\chi + \varepsilon) \quad (\text{A.12a})$$

and

$$\lim_{\varepsilon \rightarrow 0} \tilde{q}(\chi - \varepsilon) - \tilde{q}_\chi = \lim_{\varepsilon \rightarrow 0} \tilde{q}(\chi + \varepsilon) \quad (\text{A.12b})$$

respectively, where $\tilde{q}_\chi$ is the flow rate escaping from the pipe and ε is the infinitesimal distance from the reflector location χ . The proofs for Eqs. (A.12a) and (A.12b) can be made by integrating Eqs. (A.1a) and (A.1b) from $\chi - \varepsilon$ to $\chi + \varepsilon$, respectively. Evaluating Eq. (A.9) at $\chi - \varepsilon$, Eq. (A.11b) at $\chi + \varepsilon$ and substituting these two in Eq. (A.12a) gives

$$\tilde{q}_S \exp[-ik(\chi - x_S)] - \tilde{q}_R \exp[ik(\chi - x_S)] = \tilde{q}_T \exp[-ik(\chi - x_S)] \quad (\text{A.13})$$

Likewise, evaluating Eq. (A.8) at $\chi - \varepsilon$, Eq. (A.11a) at $\chi + \varepsilon$, and substituting these two in Eq. (A.12b) provide

$$\tilde{q}_S \exp[-ik(\chi - x_S)] + \tilde{q}_R \exp[ik(\chi - x_S)] - \tilde{q}_\chi = \tilde{q}_T \exp[-ik(\chi - x_S)] \quad (\text{A.14})$$

Addition of Eqs. (A.13) and (A.14) results

$$\tilde{q}_T(\omega) = \tilde{q}_S - \frac{\tilde{q}_\chi}{2} \exp[ik(\chi - x_S)] \quad (\text{A.15})$$

Substituting Eq. (A.15) in Eq. (A.14) and solving for $\tilde{q}_R$ gives

$$\tilde{q}_R(\omega) = \frac{\tilde{q}_\chi}{2} \exp[-ik(\chi - x_S)] \quad (\text{A.16})$$

The mass outflow through reflector (i.e., $\tilde{q}^\chi$) can be modeled as an orifice (Wylie et al., 1993). For steady-state, the orifice equation is given by

$$Q_0^\chi = s_L^\chi \sqrt{2gH_0^\chi} \quad (\text{A.17})$$

where H_0^χ is the steady-state pressure head at the reflector location, Q_0^χ is the steady-state flow rate extracted by the mass source, and s_L^χ is the effective cross-sectional area of the mass source. For transient state, a linearized expression of Eq. (A.17) is often used (Wang et al., 2002; Lee et al., 2006; Duan, 2017). The linearized orifice equation is given as

$$\tilde{q}^\chi = \frac{\tilde{h}^\chi}{Z_{\text{mass}}^\chi} \quad (\text{A.18})$$

wherein

$$Z_{\text{mass}}^\chi = \frac{2H_0^\chi}{Q_0^\chi} \quad (\text{A.19})$$

is the characteristic impedance of the reflector and $\tilde{h}^\chi$ is the head at the reflector location during transient state that can be obtained from Eq. (A.9). Thus, evaluating Eq. (A.9) at $x = \chi$ and substituting the resulting expression in Eq. (A.18) gives

$$\tilde{q}^\chi = \frac{Z}{Z_{\text{mass}}^\chi} \left[\tilde{q}_S \exp[-ik(\chi - x_S)] - \tilde{q}_R \exp[ik(\chi - x_S)] \right] \quad (\text{A.20})$$

Substituting Eq. (A.20) in Eq. (A.16) and solving for $\tilde{q}_R$ results

$$\tilde{q}_R = \frac{Z}{Z + 2Z_{\text{mass}}^\chi} \tilde{q}_S \exp[-2ik(\chi - x_S)], \quad (\text{A.21})$$

Substituting Eq. (A.21) in Eq. (A.9) gives

$$\tilde{h}(\omega, x|\chi) = Z\tilde{q}_S \exp[-ik(x - x_S)] + \tilde{A}_{\text{mass}}^\chi \tilde{G}(\omega, \chi), \quad (\text{A.22})$$

wherein

$$\tilde{A}_{\text{mass}}^\chi = -\frac{|Z|^2}{Z + 2Z_{\text{mass}}^\chi} \tilde{q}_S, \quad (\text{A.23a})$$

$$\tilde{G}(\omega, \chi) = \exp[-2ik(\chi - x_S)] \exp[ik(x - x_S)]. \quad (\text{A.23b})$$

A.2. Scenario II: Momentum source reflector

For a momentum point source (e.g., a drag force at a discrete blockage) which exhibits pressure head drop across the reflector, the pressure head and flow conditions are

$$\lim_{\epsilon \rightarrow 0} \tilde{h}(\chi - \epsilon) = \lim_{\epsilon \rightarrow 0} \tilde{h}(\chi + \epsilon) + \tilde{h}^\chi \quad (\text{A.24a})$$

and

$$\lim_{\epsilon \rightarrow 0} \tilde{q}(\chi - \epsilon) = \lim_{\epsilon \rightarrow 0} \tilde{q}(\chi + \epsilon) \quad (\text{A.24b})$$

respectively, where $\tilde{h}^\chi$ denotes the head loss across the reflector and can be modeled as

$$\tilde{h}^\chi = \tilde{q}^\chi Z_{\text{mom}}^\chi \quad (\text{A.25})$$

where $\tilde{q}^\chi$ can be determined using Eq. (A.8),

$$Z_{\text{mom}}^\chi = \frac{2H_0^\chi}{Q_0^\chi} = \frac{k_{HL}^\chi Q_0^\chi}{gA^2}, \quad (\text{A.26})$$

and k_{HL}^χ is the pressure head loss coefficient across the momentum source. Following the same steps as Eqs. (A.13)–(A.22) along with Eq. (A.24) yields

$$\tilde{h}(\omega, x|\chi) = Z\tilde{q}_S \exp[-ik(x - x_S)] + \tilde{A}_{\text{mom}}^\chi \tilde{G}(\omega, \chi), \quad (\text{A.27})$$

wherein

$$\tilde{A}_{\text{mom}}^\chi = \frac{Z Z_{\text{mom}}^\chi}{2Z + Z_{\text{mom}}^\chi} \tilde{q}_S \quad (\text{A.28a})$$

$$\tilde{G}(\omega, \chi) = \exp[-2ik(\chi - x_S)] \exp[ik(x - x_S)] \quad (\text{A.28b})$$

Note that both Eqs. (A.22) and (A.27) have the same form except $\tilde{A}^\chi$. Thus, for a general case where defect type is not known, Eq. (A.22) (or Eq. (A.27)) can be written as

$$\tilde{h}(\omega, x|\chi) = Z\tilde{q}_S \exp[-ik(x - x_S)] + \tilde{A}^\chi \tilde{G}(\omega, \chi) \quad (\text{A.29})$$

Appendix B. Time-reversal as optimal filter

Convolve the measured signal $\Delta H_m(t, x_R|\chi)$ with a function $g(t)$ as

$$g(t) * \Delta H_m(t, x_R|\chi) = g(t) * \Delta H(t, x_R|\chi) + g(t) * n(t) \quad (\text{B.1})$$

The objective is to determine $g(t)$ which maximizes the signal-to-noise ratio (SNR). To avoid obtaining $g(t) = \infty$, the following constraint is used: $\|g(t)\| = 1$. The SNR is defined as

$$\text{SNR} = \frac{\|g(t) * \Delta H(t, x_R|\chi)\|^2}{\mathbb{E} \|g(t) * n(t)\|^2} = \frac{\left\{ \int_{-\infty}^{\infty} g(\tau) \Delta H(t - \tau, x_R|\chi) d\tau \right\}^2}{\mathbb{E} \left\{ \int_{-\infty}^{\infty} g(\tau) n(t - \tau) d\tau \right\}^2} \quad (\text{B.2})$$

where $\|\cdot\|$ represents the $\mathbb{L}_2$ -norm and $\mathbb{E}$ is the expected value. According to Cauchy-Schwartz inequality, the numerator of Eq. (B.2) has an upper bound given by

$$\begin{aligned} & \left\{ \int_{-\infty}^{\infty} g(\tau) \Delta H(t - \tau, x_R|\chi) d\tau \right\}^2 \\ & \leq \int_{-\infty}^{\infty} g^2(\tau) d\tau \int_{-\infty}^{\infty} \Delta H^2(t - \tau, x_R|\chi) d\tau. \end{aligned} \quad (\text{B.3})$$

The denominator of Eq. (B.2) is developed as follows:

$$\begin{aligned} & \mathbb{E} \left\{ \int_{-\infty}^{\infty} g(\tau) n(t - \tau) d\tau \right\}^2 \\ & = \mathbb{E} \left\{ \int_{-\infty}^{\infty} g(\tau) n(t - \tau) d\tau \int_{-\infty}^{\infty} g(\tau') n(t - \tau') d\tau' \right\} \\ & = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(\tau) g(\tau') \mathbb{E} \{ n(t - \tau) n(t - \tau') \} d\tau d\tau' \\ & = \sigma^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(\tau) g(\tau') \delta(\tau' - \tau) d\tau' d\tau \\ & = \sigma^2 \int_{-\infty}^{\infty} g^2(\tau) d\tau \end{aligned} \quad (\text{B.4})$$

where δ is the Dirac delta function. To arrive at Eq. (B.4), the white noise assumption is invoked so that $\mathbb{E} \{ n(t - \tau) n(t - \tau') \} = \delta(\tau - \tau')$. Substituting Eqs. (B.3) and (B.4) in Eq. (B.2) gives

$$\begin{aligned} \text{SNR} & = \frac{\left\{ \int_{-\infty}^{\infty} g(\tau) \Delta H(t - \tau, x_R|\chi) d\tau \right\}^2}{\sigma^2 \int_{-\infty}^{\infty} g^2(\tau) d\tau} \\ & \leq \frac{\int_{-\infty}^{\infty} g^2(\tau) d\tau \int_{-\infty}^{\infty} \Delta H^2(t - \tau, x_R|\chi) d\tau}{\sigma^2 \int_{-\infty}^{\infty} g^2(\tau) d\tau} \\ & = \frac{\left\{ \int_{-\infty}^{\infty} g(\tau) \Delta H(t - \tau, x_R|\chi) d\tau \right\}^2}{\sigma^2 \|g(\tau)\|^2} \leq \frac{1}{\sigma^2} \int_{-\infty}^{\infty} \Delta H^2(t - \tau, x_R|\chi) d\tau \end{aligned} \quad (\text{B.5})$$

where

$$\|g(\tau)\|^2 = \int_{-\infty}^{\infty} g^2(\tau) d\tau > 0.$$

The inequality in Eq. (B.5) becomes equality, implying maximum SNR, only when $g(\tau)$ is linearly proportional to $\Delta H(t - \tau)$. That is, $g(\tau) = \alpha \Delta H(t - \tau, x_R|\chi) \forall \alpha \in \mathbb{R}$, where α is an arbitrary coefficient and often

chosen as $\|\Delta H(t - \tau, x_R | \chi)\|^{-1}$ to obtain a non-trivial solution (Turin, 1960; Krim and Viberg, 1996) such that $\|g(\tau)\| = 1$. This implies that $g(t)$ that maximizes the SNR is $\Delta H(-t) / \|\Delta H(-t)\|$. Moreover, substituting ΔH from Eq. (7) in the foregoing proposed expression of $g(t)$ gives

$$g(t) = \frac{\Delta H(-t, x_R | \chi)}{\|\Delta H(-t, x_R | \chi)\|} = \frac{G(-t, x_R | \chi)}{\|G(-t, x_R | \chi)\|} \quad (\text{B.6})$$

where $G(t) = q_{in}(t) * G(t)$ and $G(t)$ is defined in Eq. (6). Thus, the optimal estimate for $\hat{\chi}$ can be obtained by convolving Eq. (B.6) with the measurement ΔH_m and evaluating the integral at $t = 0$ as

$$\hat{\chi} = \arg \max_{\chi} B = \arg \max_{\chi} \int_{-\infty}^{\infty} \frac{G(\tau, x_R | \hat{\chi})}{\|G(\tau, x_R | \hat{\chi})\|} \cdot \Delta H_m(\tau, x_R | \chi) d\tau \quad (\text{B.7})$$

Appendix C. Estimation of defect size

For a noisy measurement $\Delta H_m(t)$, where it is assumed that noise follows Gaussian distribution with zero mean and fixed variance, the maximum likelihood estimation for defect size $\hat{A}^x$ is equivalent to the least-square solution, described as

$$\hat{A}^x = \arg \min_{A^x} \left\| \Delta \tilde{H}_m(\omega, \chi, A^x) - \Delta \tilde{H}(\omega, \hat{\chi}, \hat{A}^x) \right\|^2 \quad (\text{C.1})$$

where

$$\begin{aligned} & \left\| \Delta \tilde{H}_m(\omega, \chi, A^x) - \Delta \tilde{H}(\omega, \hat{\chi}, \hat{A}^x) \right\|^2 \\ &= \int_{-\omega_{\max}}^{\omega_{\max}} (\Delta \tilde{H}_m^C \Delta \tilde{H}_m + \Delta \tilde{H}^C \Delta \tilde{H} - 2\tilde{H}^C \tilde{H}_m) d\omega \\ &= \int_{-\omega_{\max}}^{\omega_{\max}} (\Delta \tilde{H}_m^C \Delta \tilde{H}_m + (\hat{A}^x)^2 \tilde{G}^C \tilde{G} - 2\hat{A}^x \tilde{G}^C \tilde{H}_m) d\omega \end{aligned} \quad (\text{C.2})$$

$\omega_{\max}$ is the maximum angular frequency of the injected transient signal, C denotes the complex conjugate and $\tilde{G} = \tilde{G}(\omega, \hat{\chi})$, and $\hat{\chi}$ is estimated using the time-reversal technique (i.e. Eq. (9) or Eq. (10)). Differentiating Eq. (C.2) with respect to $\hat{A}^x$ and equating it to zero gives

$$\hat{A}^x = \left(\int_{-\omega_{\max}}^{\omega_{\max}} \tilde{G}^C(\omega, \hat{\chi}) \tilde{G}(\omega, \hat{\chi}) d\omega \right)^{-1} \cdot \int_{-\omega_{\max}}^{\omega_{\max}} \tilde{G}^C(\omega, \hat{\chi}) \Delta \tilde{H}_m(\omega, \chi, A^x) d\omega \quad (\text{C.3})$$

Appendix D. Reflection coefficient of multiple defects

Consider a pipe system as shown in Fig. 2. Two reflectors in the pipeline exhibit discontinuities which split the pipe into three sections. In particular, reflector 1 at χ_1 is considered as a mass source reflector and reflector 2 at χ_2 is considered as a momentum source reflector. The pressure head perturbation $\tilde{h}(\omega, x | x_S)$ and flow rate perturbation $\tilde{q}(\omega, x | x_S)$, subjected to a mass point source at x_S , in each pipe section are given as

$$\tilde{q}(\omega, x | x_S) = \tilde{q}_S \exp[-ik(x - x_S)] + \tilde{q}_R \exp[ik(x - x_S)] \quad (\text{D.1a})$$

$$\tilde{h}(\omega, x | x_S) = Z \left(\tilde{q}_S \exp[-ik(x - x_S)] - \tilde{q}_R \exp[ik(x - x_S)] \right) \quad (\text{D.1b})$$

for Section 1;

$$\tilde{q}(\omega, x | x_S) = \tilde{c}_1 \exp[-ik(x - x_S)] + \tilde{c}_2 \exp[ik(x - x_S)] \quad (\text{D.2a})$$

$$\tilde{h}(\omega, x | x_S) = Z \left(\tilde{c}_1 \exp[-ik(x - x_S)] - \tilde{c}_2 \exp[ik(x - x_S)] \right) \quad (\text{D.2b})$$

for Section 2; where, $\tilde{c}_1$ and $\tilde{c}_2$ are the frequency-dependent amplitude of transmitted and reflected waves in Section 2, respectively,

$$\tilde{q}(\omega, x | x_S) = \tilde{c}_3 \exp[-ik(x - x_S)] \quad (\text{D.3a})$$

$$\tilde{h}(\omega, x | x_S) = Z \left(\tilde{c}_3 \exp[-ik(x - x_S)] \right) \quad (\text{D.3b})$$

for Section 3; where $\tilde{c}_3$ is the amplitude of the transmitted wave through reflector 2. Eqs. (D.1)–(D.3) are obtained in the same manner as Eqs. (A.8)–(A.11). Also, the boundary conditions for reflector 1 and reflector 2 are

$$\lim_{\epsilon \rightarrow 0} \tilde{h}(\chi_1 - \epsilon) = \lim_{\epsilon \rightarrow 0} \tilde{h}(\chi_1 + \epsilon), \quad (\text{D.4a})$$

$$\lim_{\epsilon \rightarrow 0} \tilde{q}(\chi_1 - \epsilon) - \tilde{q}^{\chi_1} = \lim_{\epsilon \rightarrow 0} \tilde{q}(\chi_1 + \epsilon), \quad (\text{D.4b})$$

$$\lim_{\epsilon \rightarrow 0} \tilde{h}(\chi_2 - \epsilon) = \lim_{\epsilon \rightarrow 0} \tilde{h}(\chi_2 + \epsilon) + \tilde{h}^{\chi_2}, \quad (\text{D.4c})$$

$$\lim_{\epsilon \rightarrow 0} \tilde{q}(\chi_2 - \epsilon) = \lim_{\epsilon \rightarrow 0} \tilde{q}(\chi_2 + \epsilon). \quad (\text{D.4d})$$

Evaluating Eq. (D.1b) at $\chi_1 - \epsilon$, Eq. (D.2b) at $\chi_1 + \epsilon$ and substituting the resulting expressions in Eq. (D.4a) gives

$$\begin{aligned} & \tilde{q}_S \exp[-ik(\chi_1 - x_S)] - \tilde{q}_R \exp[ik(\chi_1 - x_S)] \\ &= \tilde{c}_1 \exp[-ik(\chi_1 - x_S)] \\ & \quad - \tilde{c}_2 \exp[ik(\chi_1 - x_S)] \end{aligned} \quad (\text{D.5})$$

Likewise, evaluating Eq. (D.1a) at $\chi_1 - \epsilon$, Eq. (D.2a) at $\chi_1 + \epsilon$ and substituting the resulting expressions in Eq. (D.4b) gives

$$\begin{aligned} & \tilde{q}_S \exp[-ik(\chi_1 - x_S)] + \tilde{q}_R \exp[ik(\chi_1 - x_S)] - \tilde{q}^{\chi_1} \\ &= \tilde{c}_1 \exp[-ik(\chi_1 - x_S)] \\ & \quad + \tilde{c}_2 \exp[ik(\chi_1 - x_S)] \end{aligned} \quad (\text{D.6})$$

Adding Eqs. (D.5) & (D.6) and solving for $\tilde{c}_1$ gives

$$\tilde{c}_1(\omega) = \tilde{q}_S - \frac{\tilde{q}^{\chi_1}}{2} \exp[ik(\chi_1 - x_S)] \quad (\text{D.7})$$

Evaluating Eq. (D.2b) at $\chi_2 - \epsilon$, Eq. (D.3b) at $\chi_2 + \epsilon$ and substituting the resulting expressions in Eq. (D.4c) gives

$$\begin{aligned} & \tilde{c}_1 \exp[-ik(\chi_2 - x_S)] - \tilde{c}_2 \exp[ik(\chi_2 - x_S)] \\ &= \tilde{c}_3 \exp[-ik(\chi_2 - x_S)] + \frac{Z^{\chi_2}}{Z} \tilde{q}^{\chi_2} \end{aligned} \quad (\text{D.8})$$

Similarly, evaluating Eq. (D.2a) at $\chi_2 - \epsilon$, Eq. (D.3a) at $\chi_2 + \epsilon$ and substituting the resulting expressions in Eq. (D.4d) gives

$$\tilde{c}_1 \exp[-ik(\chi_2 - x_S)] + \tilde{c}_2 \exp[ik(\chi_2 - x_S)] = \tilde{c}_3 \exp[-ik(\chi_2 - x_S)] \quad (\text{D.9})$$

Adding Eqs. (D.8) & (D.9) and solving for $\tilde{c}_3$ gives

$$\tilde{c}_3 = \tilde{c}_1 - \frac{Z^{\chi_2}}{2Z} \tilde{q}^{\chi_2} \exp[ik(\chi_2 - x_S)] \quad (\text{D.10})$$

Eq. (D.10) is further simplified by substituting $\tilde{c}_1$ from Eq. (D.7) which gives

$$\tilde{c}_3 = \tilde{q}_S - \frac{1}{2} \left[\tilde{q}^{\chi_1} \exp[ik(\chi_1 - x_S)] + \frac{Z^{\chi_2}}{Z} \tilde{q}^{\chi_2} \exp[ik(\chi_2 - x_S)] \right] \quad (\text{D.11})$$

The expression for $\tilde{c}_2$ is obtained by substituting Eqs. (D.7) and (D.11) in Eq. (D.8) and given as

$$\tilde{c}_2 = -\frac{Z^{\chi_2}}{2Z} \tilde{q}^{\chi_2} \exp[-ik(\chi_2 - x_S)] \quad (\text{D.12})$$

Recall that the prime interest is to find an explicit expression for $\tilde{q}_R$ in terms of reflectors' properties which is obtained by substituting Eqs. (D.7) and (D.12) in Eq. (D.5) that gives

$$\tilde{q}_R(\omega) = \frac{\tilde{q}^{\chi_1}}{2} \exp[-ik(\chi_1 - x_S)] - \frac{Z^{\chi_2}}{Z} \frac{\tilde{q}^{\chi_2}}{2} \exp[-ik(\chi_2 - x_S)] \quad (\text{D.13})$$

It is intuitive that the reflected wave $\tilde{q}_R$ in Section 1 is composed of reflections from both reflector 1 and reflector 2. To further resolve, linearized orifice equation (i.e., Eq. (A.18)), along with Eq. (D.1b) (evaluated at χ_1) gives the expression for $\tilde{q}^{\chi_1}$ as

$$\tilde{q}^{\chi_1} = \frac{Z}{Z^{\chi_1}} \left(\tilde{q}_S \exp[-ik(\chi_1 - x_S)] - \tilde{q}_R \exp[ik(\chi_1 - x_S)] \right) \quad (\text{D.14})$$

wherein,

$$Z^{\chi_1} = \frac{2H_0^{\chi_1}}{Q_0^{\chi_1}}. \quad (D.15)$$

is the characteristic impedance of reflector 1. Similarly, the expression for $\bar{q}^{\chi_2}$ is obtained by evaluating Eq. (D.2b) at χ_2 and substituting in the linearized orifice equation (i.e. Eq. (A.18)) which gives

$$\bar{q}^{\chi_2} = \frac{2Z}{2Z + Z^{\chi_2}} [\bar{q}_S \exp[-ik(\chi_2 - x_S)] - \frac{\bar{q}^{\chi_1}}{2} \exp[ik(\chi_1 - x_S) - ik(\chi_2 - x_S)]] \quad (D.16)$$

wherein,

$$Z^{\chi_2} = \frac{2H_0^{\chi_2}}{Q_0^{\chi_2}} \quad (D.17)$$

is the characteristic impedance of reflector 2. Substituting Eqs. (D.14) and (D.16) in (D.13) and solving for $\bar{q}_R$ gives

$$\bar{q}_R = \left\{ \frac{Z}{2Z^{\chi_1}} \exp[-2ik(\chi_1 - x_S)] - \frac{Z^{\chi_2}}{(2Z + Z^{\chi_2})} \exp[-2ik(\chi_2 - x_S)] + \frac{ZZ^{\chi_2}}{2Z^{\chi_1}(2Z + Z^{\chi_2})} \exp[-2ik(\chi_2 - x_S)] \right\} \times \left[\frac{\bar{q}_S}{1 + \frac{Z}{2Z^{\chi_1}} + \frac{ZZ^{\chi_2}}{2Z^{\chi_1}(2Z + Z^{\chi_2})} \exp[2ik(\chi_1 - x_S)] \exp[-2ik(\chi_2 - x_S)]} \right] \quad (D.18)$$

The expression for $\bar{q}_R$ is comprised of information from reflector 1, information from reflector 2 and the reflector-reflector interaction (i.e., the last term in the numerator and denominator of Eq. (D.18)). For small wave reflectors, which are difficult to detect by visual inspection, $Z^{\chi} > Z$. This implies that

$$\left\{ \frac{Z}{2Z^{\chi_1}}, \frac{ZZ^{\chi_2}}{2Z + Z^{\chi_2}} \right\} \gg \frac{ZZ^{\chi_2}}{2Z^{\chi_1}(2Z + Z^{\chi_2})} \quad (D.19)$$

This relation can be explicitly seen by considering a damping free pipe system wherein $Z = a_0/(gA)$ and Eq. (D.19) which gives

$$\left\{ \frac{a_0}{gA} \gg 0 \text{ and } 4 \gg \frac{1}{2H_0^{\chi_1}} \left(\frac{a_0}{gA} Q_0^{\chi_1} \right) \right\} \quad (D.20)$$

Let $H_J^{\chi} = a_0 Q_0^{\chi}/(gA)$ be the Joukowsky's pressure change caused by the reflector, Eq. (D.20) becomes

$$\left\{ \frac{a_0}{gA} \gg 0 \text{ and } 4 \gg \frac{H_J^{\chi_1}}{2H_0^{\chi_1}} \right\} \quad (D.21)$$

From practical viewpoint, $H_0^{\chi} \gg H_J^{\chi}$, that is, the steady-state pressure at the leak location is much larger than the reflected wave amplitude due to a leak. Thus, the non-linear terms can be neglected from Eq. (D.18) which yields

$$\bar{q}_R = \frac{Z}{Z + 2Z^{\chi_1}} \bar{q}_S \exp[-2ik(\chi_1 - x_S)] - \frac{Z^{\chi_2}}{(2Z + Z^{\chi_2})} \left(1 + \frac{Z}{2Z^{\chi_1}} \right)^{-1} \bar{q}_S \exp[-2ik(\chi_2 - x_S)] \quad (D.22)$$

Since $Z/Z^{\chi} < 1$, applying the Taylor series expansion on $(1 + Z/(2Z^{\chi_1}))^{-1}$ and keeping only the first order terms of the expansion gives

$$\bar{q}_R = \frac{Z}{Z + 2Z^{\chi_1}} \bar{q}_S \exp[-2ik(\chi_1 - x_S)] + \left(-\frac{Z^{\chi_2}}{(2Z + Z^{\chi_2})} + \frac{ZZ^{\chi_2}}{2Z^{\chi_1}(2Z + Z^{\chi_2})} \right) \bar{q}_S \exp[-2ik(\chi_2 - x_S)] \quad (D.23)$$

According to Eq. (D.19), the last term is negligible in comparison with other two terms; and therefore, can be ignored. Thus, the final

expression for linearized $\bar{q}_R$ becomes

$$\bar{q}_R = \frac{Z\bar{q}_S}{Z + 2Z^{\chi_1}} \exp[-2ik(\chi_1 - x_S)] - \frac{Z^{\chi_2}\bar{q}_S}{(2Z + Z^{\chi_2})} \exp[-2ik(\chi_2 - x_S)] \quad (D.24)$$

Substituting Eq. (D.24) in Eq. (D.1b) gives the final expression for Δh as

$$\Delta \bar{h}(\omega, x|\bar{\chi}) = -\frac{|Z|^2 \bar{q}_S}{Z + 2Z^{\chi_1}} \exp[-2ik(\chi_1 - x_S) + ik(x - x_S)] + \frac{ZZ^{\chi_2} \bar{q}_S}{(2Z + Z^{\chi_2})} \exp[-2ik(\chi_2 - x_S) + ik(x - x_S)] \quad (D.25)$$

or

$$\Delta \bar{h}(\omega, x|\bar{\chi}) = \bar{A}_{\text{mass}}^{\chi_1} \bar{G}(\omega, x|\chi_1) + \bar{A}_{\text{mom}}^{\chi_2} \bar{G}(\omega, x|\chi_2) \quad (D.26)$$

where $\Delta \bar{h} = \bar{h} - Z\bar{q}_S \exp[-ik(x - x_S)]$, $\bar{A}_{\text{mass}}^{\chi_1}$ is given by Eq. (A.23a), $\bar{A}_{\text{mom}}^{\chi_2}$ is given by Eq. (A.28a) & $\bar{G}(\omega, x|\chi)$ is known by Eq. (A.23b), $\bar{\chi} = (\chi_1, \chi_2)$, $x \in [x_S, \chi_1]$ (see Fig. 2) and $\Delta \bar{h}(\omega, x|\bar{\chi})$ is the impulse response function due to both mass and momentum reflectors in the system.

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